

On the Tension Between Due Process Protection and Public Safety: The Case of an Extensive Procedural Reform in Colombia

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Abstract

We study how large procedural reforms reshape deterrence by tracing equilibrium adjustments across multiple margins of the criminal justice system. Using the staggered rollout of Colombia's comprehensive adversarial criminal procedure reform in a civil-law setting (2005-2008), we estimate both intended administrative effects and unintended effects on enforcement and crime. The reform aimed to strengthen due process protections, reduce reliance on pretrial detention, accelerate case processing, and expand early termination mechanisms. We find that the reform achieved most of its explicit objectives: pretrial detention declined by 13-28%, procedural duration fell by 19%, settlements increased by 41-65%, and adjudication rose substantially. At the same time, the reform reduced enforcement activity and case progression: arrest rates fell by 39%, and clearance rates—the share of reported crimes in which prosecutors file charges before a judge—by 16-27%. Consistent with a reduction in the certainty component of deterrence, crime increased after implementation; property crime by 29% and violent crime by 15%. These findings highlight a key implication of adversarial procedural redesign: reforms that improve internal efficiency can alter enforcement and charging behavior in ways that affect public safety. More broadly, they underscore the importance of evaluating procedural reforms as bundled institutional changes and measuring their effects along the full case-processing pipeline, from complaint to case resolution, rather than inferring mechanisms from crime outcomes alone.

Keywords: Criminal procedural reform; pretrial detention; due process protection; clearance rates; crime

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I. Introduction

The literature on how criminal law affects crime has focused primarily on how changes in substantive penal law, such as sentencing rules, incarceration, and rehabilitation policies, alter the expected costs and benefits of offending. This work has been central to understanding how the criminal justice system shapes incentives to offend through sentencing (Dominguez-Rivera et al., 2019; Owens, 2009), incapacitation (Liedka et al., 2006; Lofstrom & Raphael, 2016), and rehabilitation (Escobar et al., 2023; Gaes & Camp, 2009; Tobón, 2022).

By contrast, we know much less about how procedural law, that is, the rules governing investigation, charging, adjudication, and pretrial detention, shapes crime. Existing research examines specific procedural margins, such as higher judicial productivity (Soares & Sviatschi, 2010), due process protections (Atkins & Rubin, 2003), and reductions in pretrial detention (Cepeda-Francesc & Ramírez-Álvarez, 2023; Heaton et al., 2017; Leslie & Pope, 2017). However, procedural reforms are rarely narrow, isolated interventions. Major procedural redesigns typically bundle multiple institutional changes: reallocating authority across police, prosecutors, and judges; altering evidence production and timing constraints; and expanding or eliminating non-trial resolution pathways. These bundles can shift the likelihood that crime reports translate into arrests, charges, and convictions (certainty), pretrial detention rules and effective sanction intensity (severity), and time of adjudication (celerity) in different, and potentially offsetting, directions.

In this paper, we study the effects of criminal procedural reforms across multiple margins of adjustment of the criminal justice system. Rather than focusing on any single outcome in isolation, we evaluate the net impact of procedural change on the certainty, severity, and celerity of punishment by tracing how criminal justice actors re-optimize under new constraints and incentives. Procedural reforms can accelerate case processing, expand negotiated exits, and reduce detention, but those same changes can also shift enforcement efforts, prosecutorial screening, and the probability that reported offenses move from complaint to charge and then to conviction. Because these margins are jointly determined, the welfare-relevant effects of procedural reform cannot be inferred from any single margin; they must be evaluated by linking administrative outcomes to enforcement and crime outcomes within a unified framework.

A particularly consequential class of procedural reforms observed across many civil-law jurisdictions over the past several decades is the shift toward adversarial criminal procedure, in which oral hearings and prosecutorial-led investigations play a central role relative to traditional dossier-based processing (Damaska, 2001; Langer, 2007, 2014). These reforms are analytically useful for our purposes because they are typically bundled interventions: they change the form of adjudication (oral hearings versus written dossiers), reallocate authority across prosecutors and judges, expand early-termination mechanisms, and often alter the role and feasibility of pretrial detention. As a result, they create a

multi-margin environment in which the net deterrence effect is theoretically ambiguous and must be assessed empirically. However, crime outcomes alone are not enough to characterize the new equilibrium: the same increase (or decrease) in crime could arise from changes in punishment severity driven by heightened due process protections, or from endogenous re-optimization by prosecutors and police who alter charging, detention requests, and case selection when procedural requirements change. Tracing simultaneous changes in charging, detention requests, negotiated exits, adjudication, and enforcement activity is, therefore, necessary to interpret crime effects as equilibrium responses to procedural redesigns rather than as uninterpreted outcomes.

Despite the importance of adversarial reforms, systematic evaluations of their net effects are limited, in part because bundled procedural changes are difficult to study with credible counterfactuals. This paper addresses that gap by evaluating a major adversarial procedural transformation that is representative of the broader “adversarial revolution” in civil-law settings (Langer, 2007). Adversarial reforms are often motivated by a common set of objectives: increasing judicial productivity and reducing congestion through early-termination mechanisms (Martínez Cuéllar et al., 2008), strengthening due process protections, and reducing reliance on pretrial detention (Carranza, 2001; Duce et al., 2009; Hartmann Arboleda, 2016; McLeod, 2010) in contexts where delays and perceived impunity are salient (Dammert, 2012). At the same time, the implied deterrence effect is ambiguous *ex ante*, because the same reform can shift certainty, severity, and celerity in offsetting ways. For example, productivity gains may reflect greater prosecutorial selectivity (fewer cases pursued to charge), reducing clearance and raising incentives to offend. Alternatively, reductions in pretrial detention can lower crime in the long run but raise it in the short run (Chalfin & McCrary, 2017; Leslie & Pope, 2017). Finally, expanded early-termination mechanisms can reduce expected severity through sentence discounts while increasing the discounted cost of offending by shortening the time to conviction. The sign of the net effect is therefore an empirical question.

We study these tradeoffs through Colombia’s adversarial reform, a setting that is unusually well-suited for identifying the effects of procedural redesign as a bundle. The reform was implemented through a staggered rollout between 2005 and 2008 that divided the country into four implementation stages, allocating roughly one quarter of the population to each stage. This staged design introduced an important quasi-experimental variation in exposure during the transition, enabling credible comparisons of municipalities before and after implementation. Colombia is also a unitary country, so substantive criminal law is uniform nationwide, and penal legislation is centrally determined, limiting cross-municipality legal heterogeneity that could otherwise confound the effects of procedural change. The shift from dossier-based processing to oral hearings before an impartial judge, alongside prosecutorial-led investigations (Damaska, 2001; Langer, 2007, 2014), provides a concrete instance of *adversarialization* within a civil-law system.

We combine administrative data from prosecutors and courts with police data to measure the reform's effects along the full case-production process. On the institutional side, we track procedural timelines, charging activity (imputations), early-termination mechanisms, adjudication outcomes, and pretrial detention. On the enforcement and public-safety side, we examine arrests, clearance rates (the ratio of the number of imputations of charges and the total number of open criminal complaints), and crime rates. This integrated measurement is central to our approach because it allows us to test whether administrative “productivity” gains coincide with changes in selection and enforcement that alter the certainty of punishment, and to interpret crime responses as the net equilibrium effect of simultaneous adjustments across margins.

Our empirical strategy exploits the staggered rollout across the four implementation stages. We estimate event-study specifications with two-way fixed effects to assess pre-trends and characterize dynamic treatment effects. As a robustness check for these event studies, we use the doubly robust estimator proposed by Callaway and Sant'Anna (2021). We complement these estimates with a conditional difference-in-differences specification that uses the staged implementation to identify the causal effects of the reform on the outcomes of interest.

We find that the reform achieved most of its stated objectives: shorter post-imputation timelines, more pre-imputation settlements, higher adjudication through non-trial dispositions, and less jail-based pretrial detention, while reducing arrests and clearances and raising property and violent crime rates.

Our results show that a comprehensive procedural reform can deliver intended within-system improvements, such as shorter timelines, greater use of negotiated exits, increased adjudication through non-trial dispositions, and reduced pretrial detention, while also changing enforcement and case progression in ways that matter for public safety. This paper contributes to the law and economics literature on how criminal justice institutions shape incentives by documenting how structural procedural redesigns can shift deterrence through adjustments across the full case production process, from enforcement and charging to detention and disposition, rather than through changes in substantive criminal law alone. Our results underscore the importance of evaluating procedural reforms jointly across administrative margins rather than inferring mechanisms from crime outcomes alone.

The rest of the paper is organized as follows. Section II discusses the existing literature on criminal procedural reforms and crime. Section III explains the institutional context of the adversarial reform and its implementation in Colombia. Section IV describes the data used to estimate the impact of the adversarial reform on the administration of criminal justice, enforcement, and crime. Section V presents the empirical strategy. Section VI presents the main results, and Section VII concludes.

II. Criminal Procedure & Crime Rates

A growing literature shows that criminal procedure shapes crime even when substantive law is unchanged, by reshaping the certainty, severity, and celerity of punishment (Nagin, 2013; Nagin & Pogarsky, 2003; Pogarsky, 2002). The difficulty is that procedural reforms are rarely isolated: they relocate authority among police, prosecutors, judges, and defense, alter the production and admission of evidence, and reconfigure case-resolution pathways (Langer, 2007, 2021). Even a reform with a single stated aim, efficiency or due process, can therefore move several margins at once, leaving its net effect on crime theoretically ambiguous *ex ante*.

It is useful to organize these mechanisms as system-level outcomes that emerge after all actors re-optimize. Higher judicial productivity can raise the expected probability of punishment and deter crime (Soares & Sviatschi, 2010), but productivity gains can also reflect selection, processing fewer cases faster, so the sign depends on which offenses remain in the pipeline. Selection and substitution operate at several stages: prosecutors may charge more selectively, police may reallocate effort toward easier-to-prove offenses (Dusek, 2015), and defendants adjust bargaining as negotiated dispositions expand. Due-process-oriented reforms can simultaneously shift certainty and severity by tightening the conditions for detention and incarceration (Atkins & Rubin, 2003). Adversarial reforms are a leading case: inspired by Anglo-adversarial ideals but implemented at scale through simplified, non-trial tracks, they often concentrate discretion in the prosecutor's office, so that the courtroom-centered model governs only a small share of cases while most move through alternative routes (Langer, 2007, 2021; Lewis, 2009).

Empirical work on adversarial transitions has largely studied isolated outcomes, most often crime and arrests, rather than the full pipeline from report to disposition. The Mexican evidence is illustrative. Amuedo-Dorantes and Ibarra-Caton (2026) link the staggered accusatorial rollout to fewer arrests and more homicides, with concurrent changes in reporting and institutional trust and stronger effects under organized-crime exposure; Cepeda-Francesc and Ramírez-Álvarez (2023), using a generalized synthetic control, find higher homicides and reduced pretrial detention for property crimes in early adopters, again larger where organized crime is established, and proxy a weakened capacity to prosecute homicides; and Blanco (2012) reports reduced victimization but lower perceived security and heterogeneous shifts in trust, reporting, and investigation. Informative as these are, each is organized around crime outcomes and a limited set of intermediate margins, leaving open how the reform reshaped prosecutorial screening, charging, negotiated dispositions, and the composition of cases reaching adjudication.

Two further strands sharpen the interpretive stakes. On time horizons and heterogeneity, Hernández (2019) finds that Peru's phased reform had modest, offense-specific, and sign-varying crime effects that attenuate over time; and Cattaneo, Díaz, and Titiunik (2025) document a discrete post-reform

rise in police reports after Uruguay's reform and show, as mechanism evidence, a contemporaneous decline in imputations underscoring that early effects may reflect the bundled consequences of implementation. On adaptation under weak capacity, Hanson and Kronick (2026) show that a Venezuelan reform restricting arrests led to a sharp decline in arrests, and Magaloni and Rodríguez (2020) argue that due-process reforms reduce torture-based confessions; subsequent work by Magaloni and Salmón (2026) argues that police and prosecutors may adapt through fabricated evidence when investigative capacity remains weak. Descriptive comparative work likewise stresses that these reforms aim to curb pretrial detention and strengthen defendants' rights, yet detention policy remains contested and prone to backlash (Duce, Fuentes, & Riego, 2009).

We build on this literature by treating adversarial reform as a bundled institutional redesign and studying it in equilibrium: we document both the intended effects (procedural duration, preventive measures, settlements, and adjudication) and the unintended effects (enforcement, clearance, and crime), so that crime responses can be read against the broader re-optimization along the case trajectory. We focus on Colombia because the rollout was staggered and the administrative data allow us to track these margins jointly. The Colombian crime estimates need not generalize to every transition; our contribution is the measurement-and-interpretation framework: a joint evaluation of how police, prosecutors, judges, defendants, and complainants adjust to the new constraints, which clarifies why package-level measurement, rather than crime outcomes alone, is necessary to identify mechanisms and diagnose unintended consequences.

III. Institutional Context: The adversarial reform in Colombia

On August 31, 2004, Colombia shifted from an inquisitorial/mixed criminal justice system (Law 600 of 2000, Old Code of Criminal Procedure) to a fully adversarial system (Law 906 of 2004, the New Code of Criminal Procedure). Under Law 600 (which was in force until the end of 2004), the criminal process was dominated by a judge-centered inquiry, in which an investigating authority gathered evidence in a mostly written dossier, and the accused played a relatively passive role. Often, the prosecutor effectively acted as an investigating judge, especially in early decisions such as whether pretrial detention was necessary or whether the evidence was sufficient to proceed to trial. In contrast, Law 906 (implemented in four phases starting in early 2005) established a U.S.-style adversarial procedure with a clear separation of functions: prosecutors and the defense present evidence before an impartial judge through oral and public hearings. The reform aimed to strengthen defendants' due process rights and transparency, making the accused an active party to the proceedings and eliminating the investigative-judge role previously performed within the prosecution, while increasing efficiency through greater prosecutorial discretion and negotiated dispositions.

Since the reform transformed the entire criminal process, assessing its effects requires a clear understanding of both the previous system (Law 600) and the new adversarial system (Law 906).

We divide the analysis of the reform into three subsections to ground our empirical analysis. Figure A2 in the Online Appendix illustrates the main stages of the criminal process under Law 600 and Law 906.

a. Initiation of Investigations and Preliminary Phase

Under Law 600, a criminal case began with a confidential preliminary inquiry (*indagación preliminar*), which could be initiated *ex officio* based on any credible information about a crime. Once a suspect was identified, the prosecutor—who held both investigative and accusatory functions—formally linked the suspect to the investigation through an in-depth interrogation (*indagatoria*) and decided whether to open a formal investigation and order preventive detention. Evidence was collected in a written dossier, the defense had limited ability to intervene, and the judge or investigating prosecutor could order evidence *ex officio*, blurring the line between investigation and adjudication.

In contrast, under Law 906, a criminal report or complaint (*denuncia* or *querrela*) triggers prosecutorial action. The Prosecutor's Office (*Fiscalía*) opens a case upon receiving a criminal report (*noticia criminal*), and the preliminary investigation remains under the prosecutor's control until a formal charge is made. The investigating prosecutor no longer holds a judicial role; invasive measures such as search warrants, wiretaps, and arrests generally require authorization from a judge of guarantees (*juez de garantías*), a new role introduced by Law 906 to oversee the legality of pretrial actions.

b. Formal Charging: Imputation of Charges vs. Indictment Resolution

Under the previous law, there was no public preliminary hearing equivalent to an arraignment. After gathering sufficient evidence, the prosecutor issued an indictment resolution (*resolución de acusación*), a written decision accusing the suspect and sending the case to trial; if the evidence was insufficient, the prosecutor could decline to prosecute (*preclusión*). With few intermediate or negotiated dispositions, cases generally moved toward indictment or dismissal. Because the process was written and confidential, the accused often learned the full charges only at indictment, frequently after a lengthy investigation.

Law 906 replaced this single step with a two-step charging process. Once the prosecutor has enough evidence to reasonably suspect someone, they must appear before a judge of guarantees at an imputation hearing (*audiencia de formulación de imputación*) and formally inform the suspect of the charges and essential facts, in the presence of the judge and the defense. In U.S. terms, imputation (*formulación de imputación*) is the functional equivalent of filing or pressing charges and is distinct from the indictment (*formulación de acusación*), the later act that refers the case to trial; throughout, "imputation" denotes charging and "indictment" denotes trial referral. Imputation opens formal prosecution and judicial oversight but does not send the case to trial. Trial referral occurs only at the

indictment hearing (*audiencia de formulación de acusación*), when the case is assigned to a trial judge (*juez de conocimiento*) for the oral trial phase.

Law 906 also introduced strict procedural deadlines: after imputation, the prosecutor must file the indictment within 90 days; after indictment, the oral trial must begin within 120 days. Missed deadlines can lead to the release of the accused and a risk of procedural dismissal. Under Law 600, no equivalent deadlines existed, and defendants could remain in pretrial detention for extended periods while prosecutors completed the written file.

c. Implementation and rollout

Law 906 was implemented through a legally mandated gradual rollout across judicial districts between 2005 and 2008. The law specified the criteria for implementation and an ex-ante implementation schedule for each stage. Article 529 of Law 906/04 established that the implementation criteria include: (i) the number of offices and caseloads in the prosecutors' offices and criminal courts; (ii) the number of personnel trained in oral proceedings and projected training needed; (iii) projected demand for hearing rooms; (iv) demand for penal justice and public defenders; and (v) congestion levels. Article 530 then specifies the staged activation of the system, beginning on January 1, 2005, followed by additional districts entering on January 1, 2006, January 1, 2007, and January 1, 2008.¹

This statutory rollout structure generates variation in exposure to the reform during a short transition window; our empirical strategy leverages it to compare outcomes across districts before and after adoption. A central identification concern is whether the rollout order could be correlated with differential trends in crime or criminal justice outcomes prior to implementation, which would violate the assumption that adoption timing is exogenous to potential outcomes. Two institutional features support exogeneity. First, the schedule was legislated ex ante: Article 530 fixed the assignment of judicial districts to activation dates in 2004, before any phase began, and no district was reassigned between phases. The timing was therefore not responsive to contemporaneous trends in crime or enforcement, the scenario that most directly threatens identification in staggered designs. Second, the Article 529 criteria governing the order — prosecutor and court caseloads, trained personnel, hearing-room capacity, public-defender demand, and congestion — are measures of institutional

¹ Law 906/04, Article 529: *The following factors shall be considered for the discharge of its functions: The number of offices and cases in the Office of the General Prosecutor and in the criminal courts. The registry of officials trained in oral proceedings and the projected demand for training. The projected number of hearing rooms required. Demand for criminal justice and the need for public defense services. The level of case backlog. The rules of phased implementation are established by this law.* Law 906/04, Article 530: *Based on the analysis of the foregoing criteria, the system shall be implemented as of January 1, 2005, in the judicial districts of Armenia, Bogotá, Manizales, and Pereira. A second stage, beginning on January 1, 2006, shall include the judicial districts of Bucaramanga, Buga, Cali, Medellín, San Gil, Santa Rosa de Viterbo, and Tunja. On January 1, 2007, the judicial districts of Antioquia, Cundinamarca, Florencia, Ibagué, Neiva, Pasto, Popayán, and Villavicencio shall enter the new system. The judicial districts of Barranquilla, Cartagena, Cúcuta, Montería, Quibdó, Pamplona, Riohacha, Santa Marta, Sincelejo, and Valledupar, as well as any others that may subsequently be created, shall begin applying the system as of January 1, 2008 (authors' translation).*

readiness, not of crime outcomes; they determine which districts were operationally prepared to absorb the reform first, and any level differences they induce are absorbed by district fixed effects. We then test the observable implication of exogeneity directly: pre-implementation event-study coefficients are jointly indistinguishable from zero for arrests, clearance, and the main crime indices (see Figures 2-12B; Figures A3–A4 and Table A1 in the Online Appendix), the four waves are balanced on the socioeconomic and institutional covariates (Figure A5), and the non-retroactivity rule,² which applied the new procedure only to offenses committed after each district's effective date, further limits mechanical or anticipatory changes at the rollout boundary (Table A3).

IV. Data

We use data from four different sources to assess the intended and unintended consequences of the adversarial reform in Colombia. First, we use data from the General Prosecutor's Office (*Fiscalía General de la Nación, FGN*) on judicial decisions and judicial outcomes, for both the previous inquisitorial model information system (Prosecutorial Information System for Law 600 - *Sistema de Información Judicial Ley 600/00-SIJUF*), and the new adversarial model information system (Prosecutorial Information System for Law 906/04 - *Sistema Penal Oral Acusatorio- Ley 906 & Ley 1098-SPOA*).³ These two information systems contain case-level data since 2004 for all Colombian municipalities. From the SPOA and SIJUF, we use municipality-month information on (i) the average number of days between different procedural stages, (ii) the number of imputations of charges, (iii) the number of active cases, (iv) the number of cases with a preventive measure (house arrest or jail detention), (v) the number of settlements (before imputation), (vi) the number of convictions (at trial or guilty pleas), and (vii) the number of acquittals (at trial or agreements).

Second, the Colombian National Police (NP)⁴ provided arrest and crime data by municipality and month from 2003 to 2008.⁵ In a slightly unbalanced panel, we have information for 1,100 out of 1,122 municipalities in Colombia from 2003 to 2008.⁶ We restrict our analysis to high-impact crimes and, within that subset, select those with fewer measurement problems: homicides, assaults, muggings, business robberies, home burglaries, and vehicle thefts (De Mello et al., 2013; Di Tella &

² See the Colombian Constitutional Court Decision C-801/05 and C-1179/05.

³ With the creation of the SPOA, the adversarial reform introduced a new module in the SIEDCO, the SIDENCO (*Sistema de Denuncias y Contravenciones*). This module allowed the National Police to send crime information to the General Prosecutor's Office, integrating the SPOA and the SIEDCO information. Although the SPOA included the information registered in the SIDENCO module, the SIEDCO information only had the National Police information during our analysis period. Moreover, the new module offered a channel to exchange information between the General Prosecutor's Office and the National Police but did not change the structure or nature of the data collected by each entity.

⁴ The NP data came from the System of Statistical Information on Crime, Violations and Arrests (*Sistema de Información Estadístico, Delincuencial, Contravencional y Operativo* from the NPD-SIEDCO).

⁵ Most of the municipalities for which we do not have information for the whole period are municipalities that were part of another one and became a formal municipality at some point during the period. As these municipalities account for less than 2% of the total and are quite small, their exclusion does not constitute a threat to our results.

⁶ We restrict our study to the period between 2003 and 2008 because the SIEDCO changed in 2003, two years before the implementation of the adversarial reform (Rodríguez-Ortega et al., 2018).

Schargrodsky, 2004). These six categories account for 97% of high-impact crimes in Colombia in 2005 (Policía Nacional de Colombia-DIJIN, 2005).

Additionally, we include data on sexual and drug offenses as a limited diagnostic for potential reporting-related changes during the reform period. Sexual offenses are known to be substantially underreported and are plausibly more sensitive than other crimes to changes in reporting and trust in institutions (Briere, 1992; Scurich, 2020). For that reason, we do not treat sexual offenses as representative of other crime categories. Rather, we use them as a “high sensitivity” category: if reporting behavior were shifting sharply around the reform, it could plausibly show up more strongly in sexual offenses reports than in other crimes.⁷

In contrast, drug offenses are often characterized as victimless and are more directly shaped by policing activity and enforcement intensity than by reporting (Black, 1970; Campbell et al., 2022; Wijeratne et al., 2023). We therefore interpret drug offenses as a “low sensitivity to victim reporting” category, while recognizing that they are not immune to measurement issues because they respond mechanically to policing effort.

Accordingly, we use the reform’s estimated effects on sexual and drug offenses reports only as suggestive evidence on the plausibility that changes in reported crime reflect reporting behavior rather than underlying offending. Importantly, we do not rely on this comparison alone to rule out reporting concerns. Throughout the paper, we interpret crime effects in conjunction with changes in enforcement and case progression, since a pure reporting shift would not, by itself, predict the joint pattern of adjustments we document across these administrative margins.

To summarize crime outcomes across categories, we construct both unweighted and weighted crime indices. The unweighted index sums offenses per 100,000 inhabitants, treating all offenses equally. The weighted indices instead assign each crime category a weight equal to its average statutory sentence length (Table A2 in the Online Appendix), following Mejía, Ortega, and Ortiz (2015). The rationale is that different offenses impose different levels of social harm, so a weighted index gives greater importance to crimes that the legal system regards as more severe.⁸ We report both the unweighted index and the weighted aggregate, violent, and property crime indices in Table 9 and Figure 11.

Crime categories follow the Colombian Penal Code classification used in the underlying administrative records. Under this classification, robberies and muggings (*hurtos*) are categorized as property crimes rather than violent crimes. Accordingly, the violent crime index includes homicides and assaults, while

⁷ In Colombia, prostitution is not a crime; therefore, sexual offenses do not include prostitution or soliciting. According to official data, by 2004, the majority of sexual offense victims were minors (84.3%), and aggressors are usually well known by the victim (i.e., family members) (Medicina Legal, 2005).

⁸ Table A2 in the Online Appendix presents the weights we used for calculating the indices. We excluded sexual offenses and drug crimes from these indices since we do not have information for these crimes for 2003.

the property crime index includes muggings, business robberies, vehicle thefts, and home burglaries. Drug offenses are analyzed separately and are not included in the violent crime index. Because the aggregate crime indices include all offense categories, these classification choices affect only the disaggregated violent and property crime results.

From the NP, we also use arrest data. We normalize these crime rates per 100,000 inhabitants using yearly population projections from the Colombian National Administrative Department of Statistics (DANE). Figures A3-A4 in the Online Appendix shows the evolution of the four aggregated crime measures by stage of implementation, normalizing to one the value in the month before the implementation.⁹

Finally, we use yearly municipal-level data that includes various socioeconomic and demographic covariates from the Center for Economic Development at Universidad de Los Andes (CEDE). The CEDE panel includes information on education, income, inequality, forced displacement, and a rurality index for each municipality between 2000 and 2008. For our analysis, we use five variables identified by the literature as determinants of crime: (i) income per capita (Crutchfield, 1989; Hipp, 2007; Messner & Tardiff, 1986; Verbruggen et al., 2015); (ii) institutional capacity proxied by the municipality's fiscal performance measure (Chamlin & Cochran, 1995; Messner & Rosenfeld, 1997; Rosenfeld & Messner, 2006); (iii) the rurality index (Deller & Deller, 2011; Kowalski & Duffield, 1990; Ladbrook, 1988; Lyrly & Skipper Jr, 1981; Wells & Weisheit, 2004); (iv) education, proxied by per capita municipal expenditure on education (Buonanno & Leonida, 2006; Hjalmarsson & Lochner, 2012; Lochner, 2004; Lochner & Moretti, 2004); and (v) population density (Lobont et al., 2017; Sampson, 1983; Shichor et al., 1979, 1980). Table 1 presents descriptive statistics for the variables of interest across these four data sources.

V. Empirical Strategy

We estimate the effect of introducing the adversarial model on judicial decisions, judicial outcomes, and crime rates by exploiting the quasi-experimental variation resulting from the staggered rollout of the reform in Colombia between 2005 and 2008. Our empirical strategy consists of two alternative and complementary approaches. First, we estimate an unconditional event study model. This approach allows us to test for preexisting differences in trends (pre-trends) between treated and

⁹ All the panels show that, for all groups, most crime measures seem to increase after the implementation of the reform. The main increases seem to come from municipalities in stage two, followed by those in stages one and four. Average crime rates from stage three seem to increase in the first semester after the implementation of the reform, followed by a reduction after six to nine months, particularly for property crimes. Similarly, in Figure A3 from the Online Appendix we show the average value of the four aggregate crime measures for Colombia's five largest cities: Bogotá (stage 1), Medellín (stage 2), Cali (stage 2), Barranquilla (stage 4), Cartagena (stage 4), Cucuta (stage 4). This figure shows that, except for Medellín, the aggregate crime indices increased for this group of cities after the implementation of the procedural reform. This increase seems to be driven by an increase in property crimes and by an increase in crimes in Cali.

control (not-yet-treated) municipalities, as well as the reform's dynamic effects several months after implementation. Equation (1) represents the unconditional leads-and-lags model that we estimate:

$$Y_{i,t} = \sum_{m=-12}^{-2} \gamma_m D_{i,t+m} + \sum_{p=0}^{12} \gamma_p D_{i,t+p} + \delta_i + \mu_t + \varepsilon_{i,t} \quad (1)$$

where $Y_{i,t}$ represents the outcome variable in municipality i and month t . For judicial decisions and outcomes, $Y_{i,t}$ represents: (i) clearance rates, as the ratio of imputations to open criminal complaints; (ii) preventive measures, as the ratio of cases with preventive jail detention or house arrest to cases with imputation of charges; (iii) settlements, as the ratio between settlements and open criminal complaints; (iv) trial sentences, as the ratio of acquittals or convictions at trial over total sentences on trial, and (v) sentences, as the ratio of acquittals or convictions to open criminal complaints. Recognizing the difficulties in measuring changes in judicial and prosecutorial activity (Berggren & Gutmann, 2020; Marciano et al., 2019), we propose two types of indicators. First, rates that incorporate information delays and time spans between the denominator and numerator (CEPEJ, 2014; Marciano et al., 2019). Second, rates that control for the total number of cases when there is no logical time gap between the denominator and numerator. Further details about the construction of these variables can be found in the Online Appendix. As for crime, $Y_{i,t}$ represents: (i) unweighted crime rate index; (ii) weighted crime rate index; (iii) violent crime rate index; (iv) property crime rate index; (v) homicide rate, (vi) assault rates, (vii) sexual offenses rate; (viii) drug offenses rate, (ix) muggings rate, (x) business robberies rate, (xi) vehicle thefts rate, or (xii) home burglaries rate.

The term $\sum_{m=-12}^{-2} \gamma_m D_{i,t+m}$ denotes the sequence of lagged treatment variables ($m = -2, \dots, -12$ months, as $m = -1$ is the omitted category) and captures the potential differences in the evolution of each outcome variable during the pre-treatment period between treatment and control groups. The term $\sum_{p=0}^{12} \gamma_p D_{i,t+p}$ denotes the present and future treatment sequence, capturing the dynamic effects of the reform on the outcome variables over the 12 months after the reform was implemented.¹⁰ Moreover, we include year, month, and year-month fixed effects (μ_t) to control for national trends and possible seasonality in outcome variables. Likewise, we include municipality fixed-effects (δ_i) to control for non-observable and time-invariant municipality characteristics.¹¹ Lastly, the term $\varepsilon_{i,t}$ represents the error term, which we cluster at the judicial district and month level since the adoption of the adversarial system was assigned across judicial districts and to account for serial correlation of outcomes across municipalities within judicial districts (Abadie et al., 2023; Bertrand

¹⁰ We also estimate long-term effects using the same specification extending it to 35 months after implementation. See Figures A12 to A24 in the Online Appendix. Because the rollout is staggered, the set of districts observed at each post-implementation horizon changes over time. For this reason, we restrict primary inference to the first 12 months, where all four cohorts contribute.

¹¹ Even though the inclusion of municipality*month fixed effects or judicial districts*month fixed effects would have been more useful for identification, we do not have enough variation to estimate this large vector of fixed effects together with the effect of the reform.

et al., 2004; Cameron & Miller, 2015). In Tables A19 to A21 in the Online Appendix, we show that results are robust to using other clusters, such as at the judicial district and calendar-month level.

This specification allows us to study pre-trends between treatment and control groups. If the parameters γ_m are statistically indistinguishable from zero for $m < 0$, we would not reject the null hypothesis that $\gamma_m = 0$. Even though the parallel trends and exogeneity assumptions are not directly empirically testable, these tests suggest that using the not-yet-treated municipalities as the control group is a sensible choice and that no-anticipation effects occurred before the implementation of the adversarial reform. As our results show, these null hypotheses cannot be rejected. Nonetheless, there are still reasons to believe that municipalities in Colombia are not comparable in other observable variables, such as differences in conflict intensity, resource availability, judicial and state capacity, among other factors. Therefore, we test pre-trends among different observable characteristics, including fiscal performance, population density, forced displacement, rurality index, tax revenues, and educational investment. Figure A5 in the Online Appendix shows that the four waves of the reform were balanced on these potentially confounding variables, thus supporting the use of not-yet-treated municipalities as controls for the treated ones.¹²

In our second approach, we exploit the gradual implementation of the reform using a two-way fixed effects (TWFE) model:

$$Y_{i,t} = \beta_1 T_{i,t} + \beta_2 X_{i,t} + \delta_i + \mu_t + \varepsilon_{i,t} \quad (2)$$

where $Y_{i,t}$ represents the same set of outcome variables in municipality i and month t ; $T_{i,t}$ is a binary variable indicating whether the reform had been implemented in municipality i in month t . Vector $X_{i,t}$ contains economic, demographic, and institutional control variables, including the lag of police arrests in the municipality. Since outcome variables are measured in rates, $\widehat{\beta}_1$ should be interpreted as the average change in the outcome variable after the reform. Therefore, to obtain the relative effect of the reform, we divide this estimate by the mean of the dependent variable in the control group before the reform. We report both this mean and the implied effect in the results tables. Given the large number of outcomes in our specifications, we also test for multiple hypotheses using Westfall-Young stepdown-adjusted p-values, which we report in each table in Section VI.

In general, our results should not be interpreted as absolute changes in the outcome variables, but rather as relative changes between our treatment and control groups. Since we include municipality fixed effects and a large array of control variables, the identification of our parameter of interest

¹² In addition, we estimate a version of equation (1) in which we include a vector $X_{i,t}$ containing the mentioned economic, demographic, and institutional variables to control for time-varying municipality characteristics. With this estimation, we confirm that the parallel trends assumption is reasonable even after the inclusion of the vector of observable variables. Table A3 presents these results.

comes from changes in the outcome variables in treated municipalities relative to comparable not-yet-treated municipalities. This means that in our specification, we are not comparing remote rural areas with a substantial presence of armed groups with large urban areas. Instead, we are comparing changes across municipalities with similar levels of income per capita, institutional capacity, rurality index, education, and population density. Nonetheless, our main results are robust to different specifications.

Because our rollout is staggered, the two-way fixed-effects estimator is a weighted average of all available 2×2 difference-in-differences comparisons, and when treatment effects are heterogeneous across adoption cohorts, comparisons that use already-treated units as controls can receive negative weights and bias the estimate (Borusyak et al., 2021; Callaway & Sant’Anna, 2021; de Chaisemartin & D’Haultfoeuille, 2022; Goodman-Bacon, 2021). Rather than assume this away, we assess whether it matters in our setting. Figure A6 – A7 reports the Goodman-Bacon decomposition of our estimates, and Figures A8–A11 and Table A15 re-estimate the event studies and the difference-in-differences using the heterogeneity-robust estimators of Callaway and Sant’Anna (2021) and Wooldridge (2025), which by construction discard the problematic comparisons. These robust estimates reproduce our findings.

VI. Results

VI.I. Intended Consequences

In this section, we present our results examining the effects of the implementation of the procedural reform on several outcome variables that capture the reform’s main stated goals. The results from the estimations show that, in general, there are no significant pre-trend differences between treatment and control groups in all outcomes of interest.

i. Procedural times

Figure 1 presents procedural times before and after the reform across three stages of the penal process: (i) imputation of charges to indictment, (ii) opening of the investigation to imputation of charges, and (iii) opening of the investigation to indictment. As noted, the reform imposed strict limits on the time between imputation and the indictment hearing to protect due process and defendants’ procedural rights. These limits made prosecutors more selective in filing charges, a result we examine below, and led them to take longer before filing them.

Figure 1 shows this reallocation of time across stages. The top-left panel reports the average number of days between imputation and indictment for each of the four implementation stages from 2003 to 2010; the top-right panel plots the same evolution across all municipalities, with the reform date normalized to zero. The time between imputation and indictment fell from about 400 days before

the reform to less than 90 days afterward, consistent with the statutory maximum. At the same time, the bottom left panel shows that the time from opening the investigation to imputation increased from about 60 to about 150 days. Yet this increase was more than offset by the decline in post-imputation time: the bottom right panel shows that the total time from opening the investigation to indictment fell from about 600 days before the reform to less than 300 days one year after implementation.

The upward movement in median imputation-to-indictment duration visible in each stage immediately before its implementation date in the upper-left panel is a mechanical consequence of stock clearance under Law 600, not a behavioral anticipation of the reform. Because the non-retroactivity rule kept all pre-transition cases under Law 600 regardless of when they concluded, the months immediately preceding each district's effective date involved the closure of accumulated case backlogs, files that had been open for extended periods under the inquisitorial system's written, untimed procedure. When these long-pending cases finally reached indictment, they carried their full accumulated durations, mechanically raising the median. This reflects the resolution of inherited stock under the old regime, not a change in prosecutorial or criminal behavior. The event-study estimates in Figure 2, which use crime and enforcement outcomes dated to each district's implementation month, show flat pre-trends (Table A3), confirming the absence of anticipatory effects in the outcomes identified by the paper.

Figure 2 presents the lead-and-lags estimates from equation 1, using procedural times as the dependent variable. The figures show no statistically significant pre-treatment trend differences between treated and control groups in most periods and confirm the patterns in Figure 1: the time from opening the investigation to indictment declined by about 100 to 200 days. The bottom panel further shows that this reduction grew over time, at least during the first 12 months after implementation.

Table 2 reports the TWFE estimates from equation 2 using the three procedural-time measures as dependent variables. Columns 1–3 exclude exposure time, whereas columns 4–6 in Panel B include it. The results show that the time between imputation and indictment fell by about 229 days, a 41.2% decline relative to the pre-treatment mean. By contrast, the time from opening the investigation to imputation rose by 80 days, or more than 100%. Overall, however, the total time from opening the investigation to indictment fell by 117 days, an 18.8% decline. Panel B confirms the dynamic evolution of these timing differences shown in Figure 2.

ii. Pretrial detention

Because the adversarial reform sought to strengthen due process by raising the burden of proof for imputing charges and imposing pretrial detention, we expect jail-based pretrial detention to decline after implementation. We measure preventive measures as the number of cases with active jail-based pretrial detention, house arrest, or either measure, divided by total imputations for each crime type. Figures 3 to 5 report the lead-and-lags estimates from equation 1 for jail-based pretrial detention, house arrest, and total pretrial detention. Across crime categories, the figures show no significant pre-reform trend differences between treated and control municipalities. They also suggest a decline in jail-based pretrial detention and an increase in house arrest for homicides and drug crimes. However, post-reform estimates are imprecise, so we rely on the TWFE results.

Table 2 reports the TWFE estimates, which confirm the patterns in Figures 3 to 5. Jail-based pretrial detention falls by 33.8% for homicides, 17.7% for property crimes, 32.0% for drug offenses, and 34.9% for assaults, although the property-crime estimate is less precise and the estimate for sexual crimes is not statistically significant. House-arrest estimates are positive, smaller in absolute terms, and highly significant, but very large in percentage terms because house arrest was rarely used under the old inquisitorial system; estimated increases range from 266% for drug offenses to 859% for homicides. Pooling jail-based pretrial detention and house arrest, we find reductions across most crime categories, but only the declines for homicides, 27.5%, and drug offenses, 12.6%, are statistically significant.

iii. Pre-imputation Settlements

To improve efficiency and reduce congestion, the reform expanded negotiated solutions. Under the new system, a pre-imputation settlement hearing became mandatory for minor crimes, such as property crimes and assaults, to facilitate a reparation agreement between the alleged perpetrator and the victim so the victim withdraws the complaint before a criminal case is formally opened. We therefore expect settlement rates to increase and imputations for minor crimes to decline after implementation. We measure settlement rates as the ratio of settlements to open criminal complaints for a given crime in a municipality, allowing for information delays and time spans.

Figure 6 presents the lead-and-lags estimates for settlement rates in property crimes and assaults, the two minor-crime categories in our data. Before the reform, settlement trends did not differ significantly between treated and not-yet-treated municipalities. After implementation, settlements increased significantly, as intended, with effects growing over time for both crime categories. Table 4 reports the TWFE estimates: settlements increased by 41.7% for property crimes and 65.3% for assaults after the reform, as shown in columns 1 and 2. Columns 3 and 4 show that these increases continued to grow during the first 12 months after the reform entered into force.

iv. Adjudication rates: acquittals and convictions

We also evaluate system efficiency through adjudication rates: the share of criminal cases reaching a final decision, either in court, through a plea deal, or through a settlement. We first examine total adjudication rates, measured as acquittals or convictions. Figures 7A and 7B report the lead-and-lags estimates for acquittal and conviction rates. In the pre-treatment period, we cannot reject that treated and control municipalities followed statistically similar trends in most periods. After implementation, adjudication rates increased for almost all crime categories, both for acquittals (Figure 7A) and convictions (Figure 7B).

Table 5 reports the TWFE estimates. The reform had very large, significant effects on the share of cases reaching adjudication. For homicides, acquittals increased by 408% and convictions by 886%. The estimated increases are also large and significant for drug crimes, sexual offenses, and assaults. For property crimes, the effect on acquittals is statistically zero, but the effect on convictions is again large and significant.

These large effects have two main explanations. First, pre-reform adjudication rates were extremely low. Immediately before implementation, only 0.031% of homicide cases ended in acquittal and 0.16% in conviction; for property crimes, the corresponding rates were 0.009% and 0.093%. Even in the best-case scenarios, fewer than 1 in 200 open criminal cases reached adjudication by a judge, leaving substantial room for improvement. Second, the reform strengthened early-termination mechanisms, including plea deals, to reduce congestion by allowing sentences without trial.

To assess whether higher conviction rates partly reflect increased plea bargaining, we estimate the lead-and-lags model using the share of trial acquittals and trial convictions over total trial sentences. Figures 8A and 8B show an increase in trial acquittals and a decrease in trial convictions, as shares of total trial sentences, for homicide cases; results for other crime categories are less conclusive.

The TWFE estimates show a highly significant increase in trial acquittals as a share of total trial sentences, and a corresponding decrease in trial convictions, mainly for homicides and property crimes (Table 6). We interpret this as evidence that plea deals increased pretrial convictions and accelerated case resolution. When the prosecution has sufficient evidence, defendants may seek a plea agreement in exchange for a lesser sentence; when evidence is insufficient to secure conviction at trial, defendants may prefer not to negotiate, increasing acquittals as a share of trial judgments.

VI.II. Unintended consequences

This subsection examines three possible unintended consequences of Colombia's adversarial reform.

i. Arrest rates

The central goal of the reform was to strengthen due process by raising the procedural and evidentiary bar for arrests, imputations, and pretrial detention. Although it did not change the statutory grounds for arrest, it introduced immediate judicial review of the legality of arrests and stronger evidentiary safeguards, making procedurally or evidentially weak arrests more likely to be invalidated early.

This change could immediately alter police incentives. Once Law 906 became operative in a judicial district, arrests were subject to the adversarial system's requirements, making their expected payoff more dependent on evidentiary compliance and prosecutorial timelines. Officers may therefore have avoided marginal arrests that were unlikely to survive downstream review, thereby producing an abrupt decline in arrest rates.

Figure 9 reports lead-and-lags estimates for police arrests per 100,000 inhabitants. Pre-treatment trends between treated and control municipalities are statistically similar in most periods. After implementation, arrest rates fell sharply and significantly. Table 7 shows a 39% reduction that increases over the first 12 months after implementation.

ii. Clearance rates

The reform could reduce clearance rates, defined as imputations divided by open criminal complaints for each crime category, through two channels. First, mandatory pre-imputation settlement hearings encouraged closing minor complaints before a formal penal case; as Table 4 shows, agreements between victims and accused increased, implying fewer imputations. Second, by limiting the time between imputation and indictment, the reform likely made prosecutors more selective, filing charges mainly when evidence was sufficient to reach indictment within the new deadlines. Consistent with this, Figure 1's lower-left panel shows investigation-to-imputation time increased after implementation.

Figure 10 tests this prediction using clearance rates as the dependent variable. Before the reform, trends did not differ significantly for most crime categories, except assaults. After implementation, clearance rates fell significantly across all categories. TWFE estimates in Table 8 confirm reductions of 23% for homicides, 24% for property crimes, 16% for drug offenses, 25% for sexual offenses, and 27% for assaults. For property crimes, the data cannot separate the effects of settlements from procedural time limits. For homicides, drug offenses, and sexual offenses, where pre-imputation

settlements are legally unavailable, the estimates likely capture the effect of procedural time limits, complementing Figure 1 and Table 2's evidence of longer investigation-to-imputation times.

iii. Crime rates

We finally estimate the reform's overall effect on crime rates. As discussed in Sections I and II, the reform altered the certainty, severity, and celerity of punishment in opposing directions: some changes weakened deterrence, while others strengthened it. The net effect on crime is therefore theoretically ambiguous and must be estimated empirically. Figure 11 reports the lead-and-lags estimates for four indices: the unweighted crime index, weighted aggregate crime index, weighted violent crime index, and weighted property crime index. In the 12 months before implementation, treated and not-yet-treated municipalities followed statistically similar trends across all four indices. After implementation, all four indices increased sharply.

Table 9 corroborates these results using TWFE specifications. The reform increased the unweighted crime index by 30%, the weighted aggregate index by 19%, the violent crime index by 15%, and the property crime index by 29%; all estimates are highly significant. Adding exposure time in columns 5–8 shows that effects grew during the first 12 months after implementation. After 12 months, the estimated increases are 45% for the unweighted index, 27% for the weighted index, 16% for the violent crime index, and 58% for the property crime index.

These magnitudes are comparable to those in related studies. Cepeda-Francesse and Ramírez-Álvarez (2023) find that Mexico's adversarial reform increased homicide rates by 7%, while Cattaneo et al. (2025) estimate that police reports in Montevideo rose by about 25 reports per day, or 8.2%, in the first week after implementation. Disaggregating the indices, the increase in violent crime is driven by assaults, with no significant change in homicides or sexual offenses (Figure 12A; Table 10A). Within property crime, all four categories—muggings, business robberies, vehicle thefts, and home burglaries—increase once municipality controls are included, with the largest and most precisely estimated effects for muggings, business robberies, and home burglaries (Figure 12B; Table 10B).

VI.III. Robustness Checks

Our results on crime are highly robust to different specifications. We estimate a highly flexible difference-in-differences model (Wooldridge, 2025) to account for some of the issues in the staggered TWFE specification (See Table A15 in the Online Appendix). We conduct two additional falsification tests to further assess the robustness of our results. First, in Table A3, we present the results of the conditional event study during the pre-treatment period (12 months before implementation). The results indicate that parallel trends hold. Second, we randomly assign all Colombian municipalities to different implementation stages and re-estimate equation (1). We repeat this random assignment 100

times. Table A5 includes the average coefficients and standard errors across these 100 estimations. Third, we preserve the geographic cohort structure of the actual rollout and assign placebo implementation dates to the first three cohorts within the available pre-reform period, while the fourth cohort remains untreated and serves as the comparison group. See Tables A6 to A8. We do not observe any effect of these counterfactual reforms on crime.

Our results also hold when we exclude Bogotá, Medellín, and Cali, Colombia's three largest cities, (Tables A9 to A11). Moreover, we estimate the model excluding the lag of arrest (Table A4) and excluding the whole vector of covariates $X_{i,t}$, obtaining similar results (Tables A12 to A14). Finally, due to the large number of zeros in our dependent variables, we follow Chen and Roth (2024) and estimate our main model using a Poisson specification with the number of crimes of a given type in municipality i in time t ($Y_{i,t}$) as the dependent variable, while including the same right-hand side variables and additional control for the municipality's population (Tables A16 to A18).

VII. Concluding remarks

We evaluate Colombia's 2005–2008 transition from an inquisitorial (Law 600/2000) to an adversarial system (Law 906/2004) as a bundled institutional redesign rather than a single-margin policy change, using the legally mandated staggered rollout across judicial districts together with administrative data from both procedural regimes and police data on arrests and crime. This design lets us trace the reform's effects along the full case-production pipeline, from complaint to disposition.

The reform largely achieved its stated objectives. Post-imputation procedural duration fell sharply (the imputation-to-indictment interval dropped 41.2%, and the investigation-to-indictment interval 18.8%); jail-based pretrial detention declined across most offenses (33.8% homicides, 17.7% property, 32.0% drug, 34.9% assaults), with domiciliary detention rising but not fully offsetting, so total pretrial detention fell for homicides (27.5%) and drug offenses (12.6%) and was roughly unchanged for other crimes; settlements rose for minor offenses (41.7% property, 65.3% assaults); and adjudication increased, driven by non-trial dispositions rather than trial convictions.

These gains coexisted with unintended consequences for enforcement and case progression. Arrest rates fell by about 39% (roughly 52% within 12 months under exposure-time specifications), and clearance rates (imputations relative to reported crimes) fell across all major categories (about 23% for homicides, 24% for property, 16% for drugs, 25% for sexual offenses, 27% for assaults). For minor offenses, the decline is consistent with the new pre-imputation settlement hearings diverting cases from formal charging; for offenses where settlements are not permitted (homicides, drug, sexual), it points to heightened selectivity at charging under tighter timelines. Consistent with a fall in the certainty of punishment, the weighted aggregate crime index rose 18.7%, the violent index 15.2%, and the property index 28.7%, with effects growing over the first post-reform year.

More broadly, the results travel beyond the Colombian adversarial reform. The central lesson is that comprehensive procedural reforms can succeed on their internal administrative objectives while still weakening deterrence through equilibrium responses that reduce enforcement and case progression. This is not a “due process versus public safety” trade-off: strengthening fairness and curbing excessive detention are core objectives of modern criminal justice, and the implications are institutional. Where tighter timelines, higher evidentiary demands, and expanded negotiated exits operate under binding investigative and prosecutorial capacity, the system may rationally screen out cases and reduce enforcement, precisely the margin that most affects deterrence. Reforms should therefore be designed and evaluated by measuring effects across the whole pipeline and by pairing rights-protective procedures with the capacity needed to continue converting reports into credible charges, rather than inferring mechanisms from crime outcomes alone.

References

- Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. (2023). When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, 138(1), 1-35.
- Amuedo-Dorantes, C., & Ibarra-Caton, M. (2026). How Mexican judicial reforms may have fueled crime: Arrest trends and trust erosion. *Criminology & Public Policy*, 25(1), 89-112.
- Atkins, R. A., & Rubin, P. H. (2003). Effects of criminal procedure on crime rates: Mapping out the consequences of the exclusionary rule. *The Journal of Law and Economics*, 46(1), 157–179.
- Berggren, N., & Gutmann, J. (2020). Securing personal freedom through institutions: The role of electoral democracy and judicial independence. *European Journal of Law and Economics*, 1–22.
- Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *The Quarterly journal of economics*, 119(1), 249-275.
- Black, D. J. (1970). Production of crime rates. *American Sociological Review*, 733–748.
- Blanco, L. R. (2012). The impact of reform on the criminal justice system in Mexico.
- Borusyak, K., Jaravel, X., & Spiess, J. (2021). Revisiting event study designs: Robust and efficient estimation. *arXiv Preprint arXiv:2108.12419*.
- Briere, J. (1992). Methodological issues in the study of sexual abuse effects. *Journal of Consulting and Clinical Psychology*, 60(2), 196.
- Buonanno, P., & Leonida, L. (2006). Education and crime: Evidence from Italian regions. *Applied Economics Letters*, 13(11), 709–713.
- Callaway, B., & Sant’Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.
- Cameron, A. C., & Miller, D. L. (2015). A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372.
- Campbell, W., Griffiths, E., & Hinkle, J. (2022). The behavior of police: Class, race, and discretion in drug enforcement. *Police Practice and Research*, 23(3), 337–354.
- Carranza, E. (2001). *Justicia penal y sobrepoblación penitenciaria: Respuestas posibles*. Siglo XXI.
- Cattaneo, M. D., Díaz, C., & Titiunik, R. (2025). Randomization inference for before-and-after studies with multiple units: an application to a criminal procedure reform in Uruguay. *Journal of the Royal Statistical Society Series A: Statistics in Society*, qnaf192.
- Cepeda-Francesse, C. A., & Ramírez-Álvarez, A. A. (2023). Reforming justice under a security crisis: The case of the criminal justice reform in Mexico. *World Development*, 163, 106148.
- CEPEJ. (2014). *European Judicial Systems: Efficiency and Quality of Justice* [Technical Report]. European Commission for the Efficiency of Justice. <http://www.astrid-online.it/static/upload/cepe/cepej-study-23-report-en-web.pdf>
- Chalfin, A., & McCrary, J. (2017). Criminal deterrence: A review of the literature. *Journal of Economic Literature*, 55(1), 5–48.
- Chamlin, M. B., & Cochran, J. K. (1995). Assessing Messner and Rosenfeld’s institutional anomie theory: A partial test. *Criminology*, 33(3), 411–429.
- Chen, J., & Roth, J. (2024). Logs with zeros? Some problems and solutions. *The Quarterly Journal of Economics*, 139(2), 891-936.
- Consejo de la Judicatura Federal. (2023). *El ABC del Sistema de Justicia Penal Adversarial*. <https://www.cjf.gob.mx/sjpa/>
- Crutchfield, R. D. (1989). Labor stratification and violent crime. *Social Forces*, 68(2), 489–512.
- Damaska, M. (2001). Models of criminal procedure. *Zbornik PFZ*, 51, 477.

- Dammert, L. (2012). *Fear and crime in Latin America: Redefining state-society relations* (Vol. 3). Routledge.
- de Chaisemartin, C., & D'Haultfoeuille, X. (2022). *Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey*. National Bureau of Economic Research.
- De Mello, J., Mejía, D., & Suárez, L. (2013). The pharmacological channel revisited: Alcohol sales restrictions and crime in Bogotá. *Documento CEDE*, 2013–20.
- Deller, S. C., & Deller, M. W. (2011). Structural shifts in select determinants of crime with a focus on rural and urban differences. *Criminology, Criminal Justice, Law & Society*, 12(3), 120.
- Di Tella, R., & Schargrodsky, E. (2004). Do police reduce crime? Estimates using the allocation of police forces after a terrorist attack. *American Economic Review*, 94(1), 115–133.
- Dominguez-Rivera, P., Lofstrom, M., & Raphael, S. P. (2019). *The effect of sentencing reform on crime rates: Evidence from California's Proposition 47*.
- Duce, M., Fuentes, C., & Riego, C. (2009). La reforma procesal penal en América Latina y su impacto en el uso de la prisión preventiva. *Prisión Preventiva y Reforma Procesal Penal En América Latina. Evaluación y Perspectivas*. CEJA-JSCA, 13–73.
- Dušek, L. (2015). Time to punishment: The effects of a shorter criminal procedure on crime rates. *International Review of Law and Economics*, 43, 134–147.
- Escobar, M. A., Tobón, S., & Vanegas-Arias, M. (2023). Production and persistence of criminal skills: Evidence from a high-crime context. *Journal of Development Economics*, 160, 102969.
- Gaes, G. G., & Camp, S. D. (2009). Unintended consequences: Experimental evidence for the criminogenic effect of prison security level placement on post-release recidivism. *Journal of Experimental Criminology*, 5(2), 139–162.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.
- Hanson, R., & Kronick, D. (2026). Official vigilantism. *World Development*, 198, 106978
- Hartmann Arboleda, Mi. (2016). *La detención preventiva y la reforma procesal penal en Colombia*.
- Heaton, P., Mayson, S., & Stevenson, M. (2017). The downstream consequences of misdemeanor pretrial detention. *Stan. L. Rev.*, 69, 711.
- Hernández, W. (2019). Do criminal justice reforms reduce crime and perceived risk of crime? A quasi-experimental approach in Peru. *International Review of Law and Economics*, 58, 89–100.
- Hipp, J. R. (2007). Income inequality, race, and place: Does the distribution of race and class within neighborhoods affect crime rates? *Criminology*, 45(3), 665–697.
- Hjalmarsson, R., & Lochner, L. (2012). The impact of education on crime: International evidence. *CESifo DICE Report*, 10(2), 49–55.
- Kowalski, G. S., & Duffield, D. (1990). The Impact of the Rural Population Component on Homicide Rates in the United States: A County-Level Analysis. *Rural Sociology*, 55(1), 76–90.
- Ladbrook, D. A. (1988). Why are crime rates higher in urban than in rural areas?—Evidence from Japan. *Australian & New Zealand Journal of Criminology*, 21(2), 81–103.
- Langer, M. (2007). Revolution in Latin American criminal procedure: Diffusion of legal ideas from the periphery. *The American Journal of Comparative Law*, 55(4), 617–676.
- Langer, M. (2014). The Long Shadow of the Adversarial and Inquisitorial Categories. In *The Oxford Handbook of Criminal Law*.
- Langer, M. (2021). Plea bargaining, conviction without trial, and the global administratization of criminal convictions. *Annual Review of Criminology*, 4(1), 377–411.
- Leslie, E., & Pope, N. G. (2017). The unintended impact of pretrial detention on case outcomes: Evidence from New York City arraignments. *The Journal of Law and Economics*, 60(3), 529–557.

- Lewis, M. K. (2009). Taiwan's new adversarial system and the overlooked challenge of efficiency-driven reforms. *Virginia Journal of International Law*, 49(3).
- Liedka, R. V., Piehl, A. M., & Useem, B. (2006). The crime-control effect of incarceration: Does scale matter? *Criminology & Public Policy*, 5(2), 245–276.
- Listokin, Y. (2003). Does more crime mean more prisoners? An instrumental variables approach. *The Journal of Law and Economics*, 46(1), 181–206.
- Lobont, O.-R., Nicolescu, A.-C., Moldovan, N.-C., & Kuloğlu, A. (2017). The effect of socioeconomic factors on crime rates in Romania: A macro-level analysis. *Economic Research-Ekonomska Istraživanja*, 30(1), 91–111.
- Lochner, L. (2004). Education, work, and crime: A human capital approach. *International Economic Review*, 45(3), 811–843.
- Lochner, L., & Moretti, E. (2004). The effect of education on crime: Evidence from prison inmates, arrests, and self-reports. *American Economic Review*, 94(1), 155–189.
- Lofstrom, M., & Raphael, S. (2016). Incarceration and crime: Evidence from California's public safety realignment reform. *The ANNALS of the American Academy of Political and Social Science*, 664(1), 196–220.
- Lyerly, R. R., & Skipper Jr, J. K. (1981). Differential rates of rural-urban delinquency: A social control approach. *Criminology*, 19(3), 385–399.
- Magaloni, B., & Salmón, E. (2026). Fabricated justice: How due process reform enables evidence manipulation. *World Development*, 199, 107222.
- Magaloni, B., & Rodriguez, L. (2020). Institutionalized police brutality: Torture, the militarization of security, and the reform of inquisitorial criminal justice in Mexico. *American Political Science Review*, 114(4), 1013–1034.
- Marciano, A., Melcarne, A., & Ramello, G. B. (2019). The economic importance of judicial institutions, their performance and the proper way to measure them. *Journal of Institutional Economics*, 15(1), 81–98.
- Martínez Cuéllar, M., Hernández Luna, Y., & Parra González, A. del P. (2008). *Boletín Dirección de Justicia y Seguridad: Cifras de Justicia, Jurisdicción Penal 1996-2007*. DNP.
https://colaboracion.dnp.gov.co/CDT/Prensa/Publicaciones/14-20_01_09_cifras_justicia.pdf
- McLeod, A. M. (2010). Exporting US Criminal Justice. *Yale L. & Pol'y Rev.*, 29, 83.
- Medicina Legal. (2005). *Delito Sexual*.
<https://www.medicinalegal.gov.co/documents/20143/49490/Delito+Sexual.pdf>
- Mejía, D., Ortega, D., & Ortiz, K. (2015). Un análisis de la criminalidad urbana en Colombia. *Consultado en www.caf.com*.
- Messner, S. F., & Rosenfeld, R. (1997). Political restraint of the market and levels of criminal homicide: A cross-national application of institutional-anomie theory. *Social Forces*, 75(4), 1393–1416.
- Messner, S. F., & Tardiff, K. (1986). Economic inequality and levels of homicide: An analysis of urban neighborhoods. *Criminology*, 24(2), 297–316.
- Ministerio Público Peru. (2005). *Plan de Implementación del Nuevo Código Procesal Penal*.
<https://cdn.www.gob.pe/uploads/document/file/1611021/PLAN-DE-IMPLEMENTACION-DEL-NUEVO-CPP.pdf.pdf>
- Nagin, D. S. (2013). Deterrence: A review of the evidence by a criminologist for economists. *Annual Review of Economics*, 5(1), 83–105.
- Nagin, D. S., & Pogarsky, G. (2003). An experimental investigation of deterrence: Cheating, self-serving bias, and impulsivity. *Criminology*, 41(1), 167–194.
- Ortega, E. (2016, May 4). Sólo 5 estados aplican el nuevo sistema penal. *El Financiero*.
<https://www.elfinanciero.com.mx/nacional/solo-estados-aplican-nuevo-sistema-penal/>

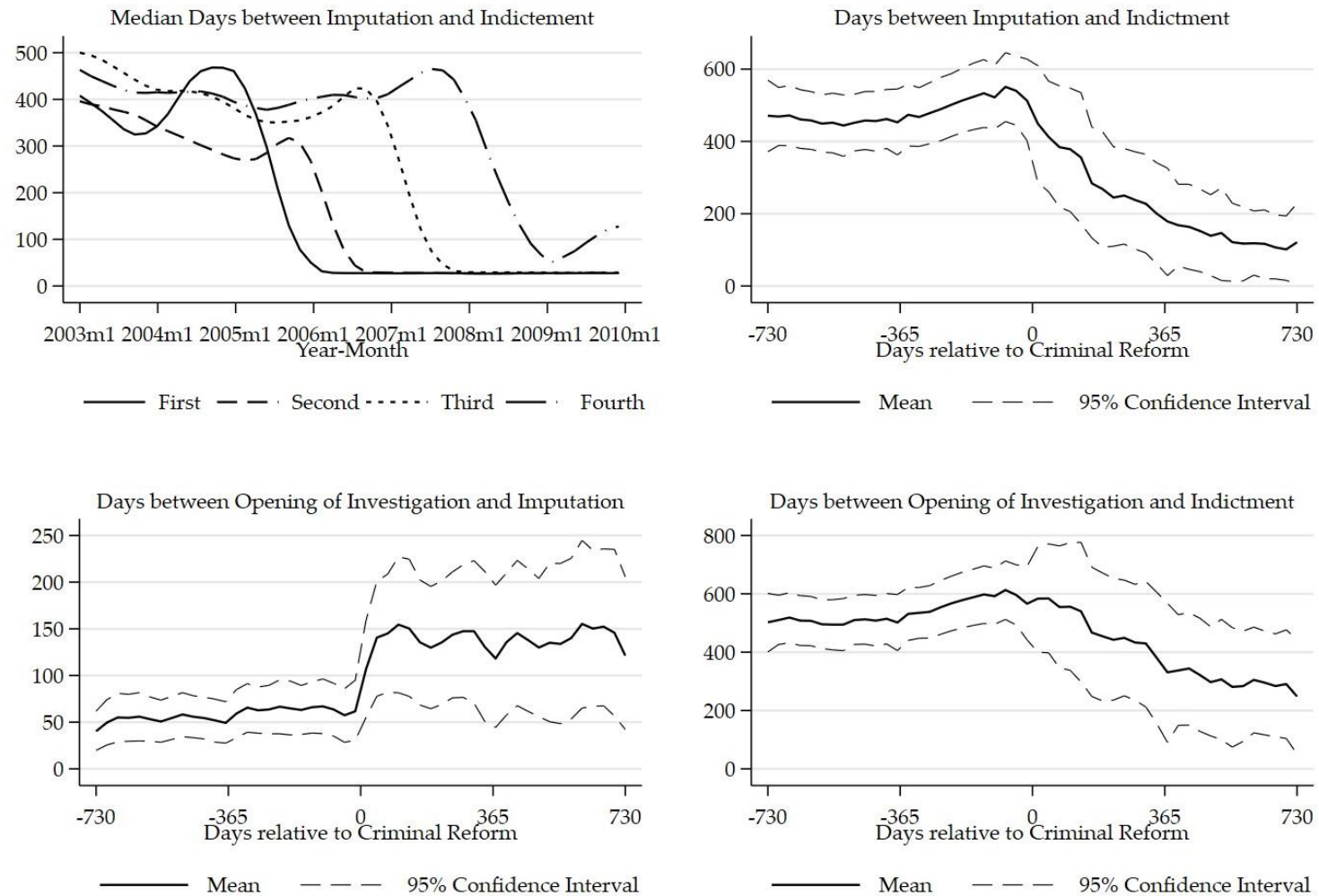
- Owens, E. G. (2009). More time, less crime? Estimating the incapacitative effect of sentence enhancements. *The Journal of Law and Economics*, 52(3), 551–579.
- Pfaff, J. F. (2008). The empirics of prison growth: A critical review and path forward. *The Journal of Criminal Law and Criminology*, 547-619.
- Poder Judicial. (2022). *Comunicado Unidad Equipo Técnico Insitucional del Código Procesal Penal*. https://www.pj.gob.pe/wps/wcm/connect/NCPP/s_ncpp/as_info/
- Pogarsky, G. (2002). Identifying “deterrable” offenders: Implications for research on deterrence. *Justice Quarterly*, 19(3), 431–452.
- Polícia Nacional de Colombia-DIJIN. (2005). *Delitos de Impacto* (pp. 14–25).
- Rodríguez-Ortega, J. D., Mejía-Londoño, D., Caro-Zambrano, L. del P., Romero-Hernández, M., & Campos-Méndez, F. (2018). Implicações do processo de integração dos registros administrativos da criminalidade entre o SPOA da Fiscalia Geral e o SIEDCO da Policia Nacional da Colômbia, e a implementação do aplicativo “¡ADenunciar!” sobre as cifras de criminalidade. *Revista Criminalidad*, 60(3), 9–27.
- Rosenfeld, R., & Messner, S. F. (2006). The origins, nature and prospects of institutional-anomie theory. *The Essential Criminology Reader*, 164–173.
- Rosenfeld, R., & Wallman, J. (2019). Did de-policing cause the increase in homicide rates? *Criminology & Public Policy*, 18(1), 51-75.
- Sampson, R. J. (1983). Structural density and criminal victimization. *Criminology*, 21(2), 276–293.
- Scurich, N. (2020). Introduction to this special issue: Underreporting of sexual abuse. *Behavioral Sciences & the Law*, 38(6), 537–656.
- Shichor, D., Decker, D. L., & O'BRIEN, R. M. (1979). POPULATION DENSITY AND CRIMINAL VICTIMIZATION Some Unexpected Findings in Central Cities. *Criminology*, 17(2), 184–193.
- Shichor, D., Decker, D. L., & O'Brien, R. M. (1980). The relationship of criminal victimization, police per capita and population density in twenty-six cities. *Journal of Criminal Justice*, 8(5), 309–316.
- Soares, Y., & Sviatschi, M. M. (2010). Does court efficiency have a deterrent effect on crime? Evidence for Costa Rica. *Unpublished Manuscript*.
- Tobón, S. (2022). Do better prisons reduce recidivism? Evidence from a prison construction program. *Review of Economics and Statistics*, 104(6), 1256–1272.
- Verbruggen, J., Apel, R., Van der Geest, V. R., & Blokland, A. A. (2015). Work, income support, and crime in the Dutch welfare state: A longitudinal study following vulnerable youth into adulthood. *Criminology*, 53(4), 545–570.
- Wells, L. E., & Weisheit, R. A. (2004). Patterns of rural and urban crime: A county-level comparison. *Criminal Justice Review*, 29(1), 1–22.
- Wijeratne, A. K., Ravikumar, N., Bandara, P. M., & Kuhaneswaran, B. (2023). Prognostication of Crime Using Bagging Regression Model: A Case Study of London. In *Handbook of Research on Technological Advances of Library and Information Science in Industry 5.0* (pp. 462–478). IGI Global.
- Wooldridge, J. (2025). Two-way fixed effects, the two-way mundlak regression, and difference-in-differences estimators. *Empirical Economics*, 69, 2545-2587.
- Zorro Medina, A. (2019). Adversariality, Plea Bargaining, and Prison Population Growth: Evidence from a Natural Experiment. *Available at SSRN 3686925*.
- Zorro Medina, A. (2020). *The Failed War on Pretrial Detention: Evidence from a Quasi-Experimental Reform* [Doctoral Dissertation]. Yale University.

Table 1. Descriptive Statistics Variables in Main Results

Variables	Observations	Mean	Std. Dev	Min	Max
Unweighted Crime	78,894	14.99	23.58	0	834.33
Crime Index	78,894	2.57	4.19	0	134.42
Violent Crime Index	78,894	4.11	8.34	0	307.25
Property Crime Index	78,894	1.37	3.03	0	93.73
Homicides	78,894	3.76	9.76	0	374.90
Assaults	78,894	5.29	12.57	0	834.33
Sexual Offenses	62,977	4.49	8.44	0	232.29
Drug Offenses	62,977	3.51	9.77	0	941.37
Muggings	78,894	2.92	8.16	0	286.81
Business Robberies	78,894	0.96	3.51	0	99.50
Vehicle Thefts	78,894	0.43	2.30	0	183.53
Home Burglaries	78,894	1.63	5.68	0	191.94
Arrests	78,894	1.52	1.53	0	6.77
% Rural Population	78,894	1.05	0.65	0	4.09
Education Investment	77,660	3.55	0.94	0	12.34
Industry Tax Revenue	78,810	0.01	0.03	0	0.52
Fiscal Performance	77,790	4.08	0.16	2.52	4.51
Population Density	78,894	3.85	1.23	0.14	9.58
Forced Displacement Expulsion	78,918	3.83	2.11	0	9.63
Forced Displacement Reception	78,918	3.07	2.24	0	10.61

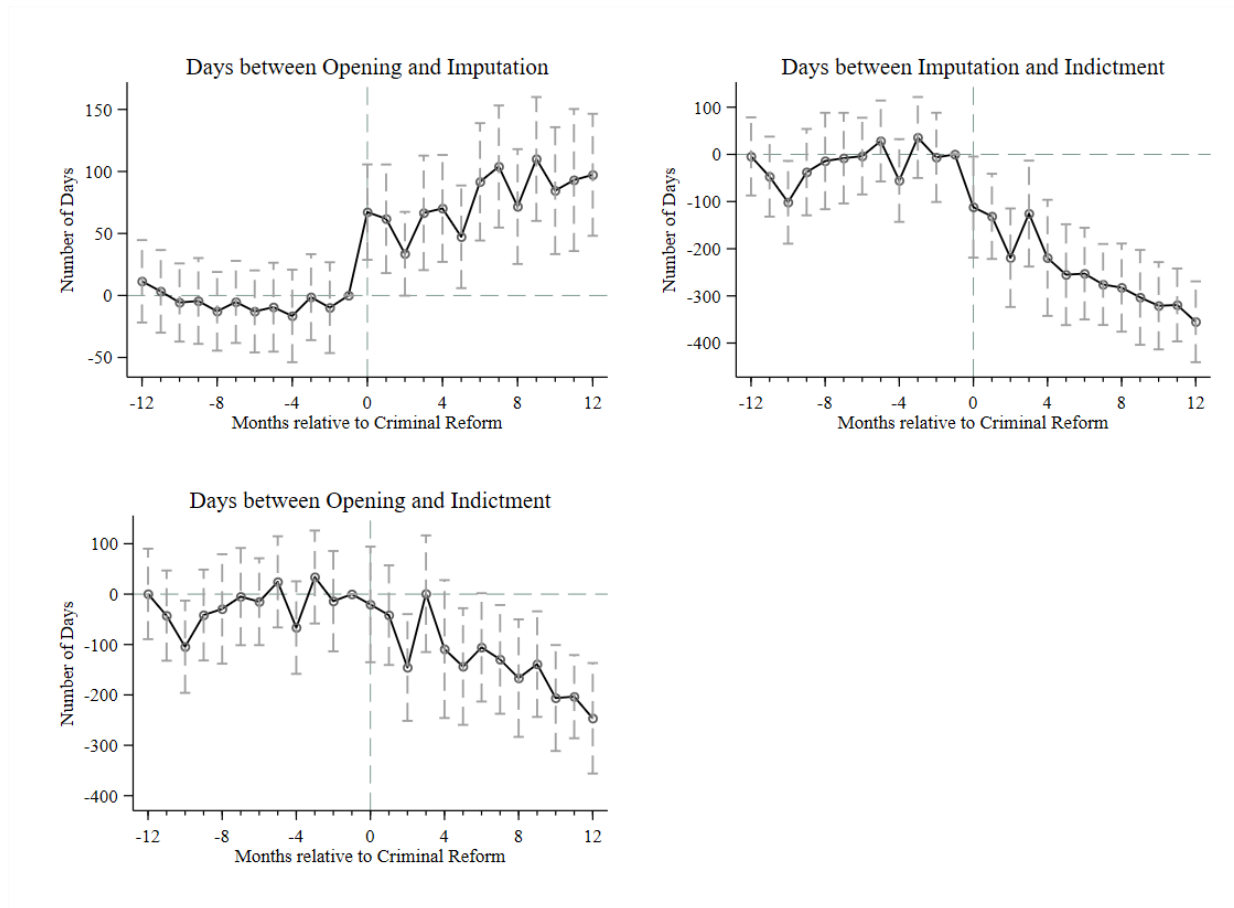
Note: this table shows the number of observations, mean, standard deviation, minimum and maximum for our variables of interests, both result and control. Crimes and arrests are reported per 100,000 Inhabitants.

Figure 1. Procedural Length Reduction in Colombia after the Reform



Note: this figure shows the evolution of the time between different stages of the criminal process. The top-left panel shows the evolution over time of the median number of days between imputation and indictment by stage of implementation of the reform, while the other three panels show the evolution of the average of other measures across all municipalities over the time relative to the date of the implementation of the reform, including the 95% confidence intervals represented by the dashed lines.

Figure 2. Leads-and-lags model: Procedural Times



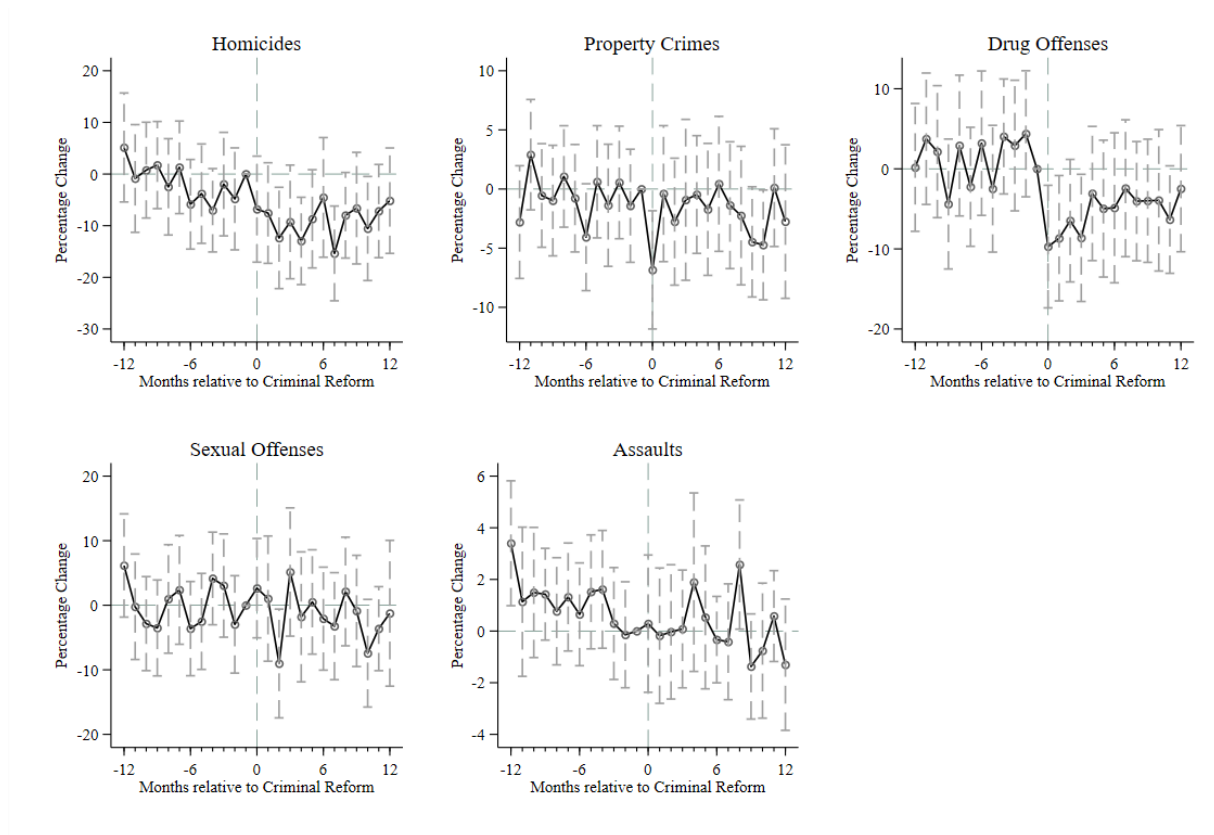
Note: this figure shows the results of an event study of the number of days between different stages of the criminal process as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors.

Table 2. Difference-in-Difference Results for Procedural Times

VARIABLES Days between	Panel A			Panel B		
	(1) Imputation - Indictment	(2) Opening - Imputation	(3) Opening - Indictment	(4) Imputation - Indictment	(5) Opening - Imputation	(6) Opening - Indictment
T	-229.14*** (14.62)	79.84*** (5.78)	-117.11*** (15.42)	-197.86*** (14.19)	84.68*** (6.45)	-68.18*** (15.12)
Exposure Time to T				-7.92*** (0.96)	-1.26*** (0.38)	-11.50*** (0.97)
Constant	504.66*** (82.60)	37.01 (34.54)	468.04*** (95.09)	491.36*** (77.28)	34.97 (34.71)	441.37*** (84.55)
Observations	17,842	17,750	17,526	17,842	17,750	17,526
R-squared	0.39	0.26	0.33	0.391	0.260	0.339
Year Month & Month- Year FE	YES	YES	YES	YES	YES	YES
Municipio FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
P-wyong P-Value	0.00	0.00	0.00	0.00	0.00	0.00
Mean T = 0	556.67	66.81	622.60	556.67	66.81	622.60

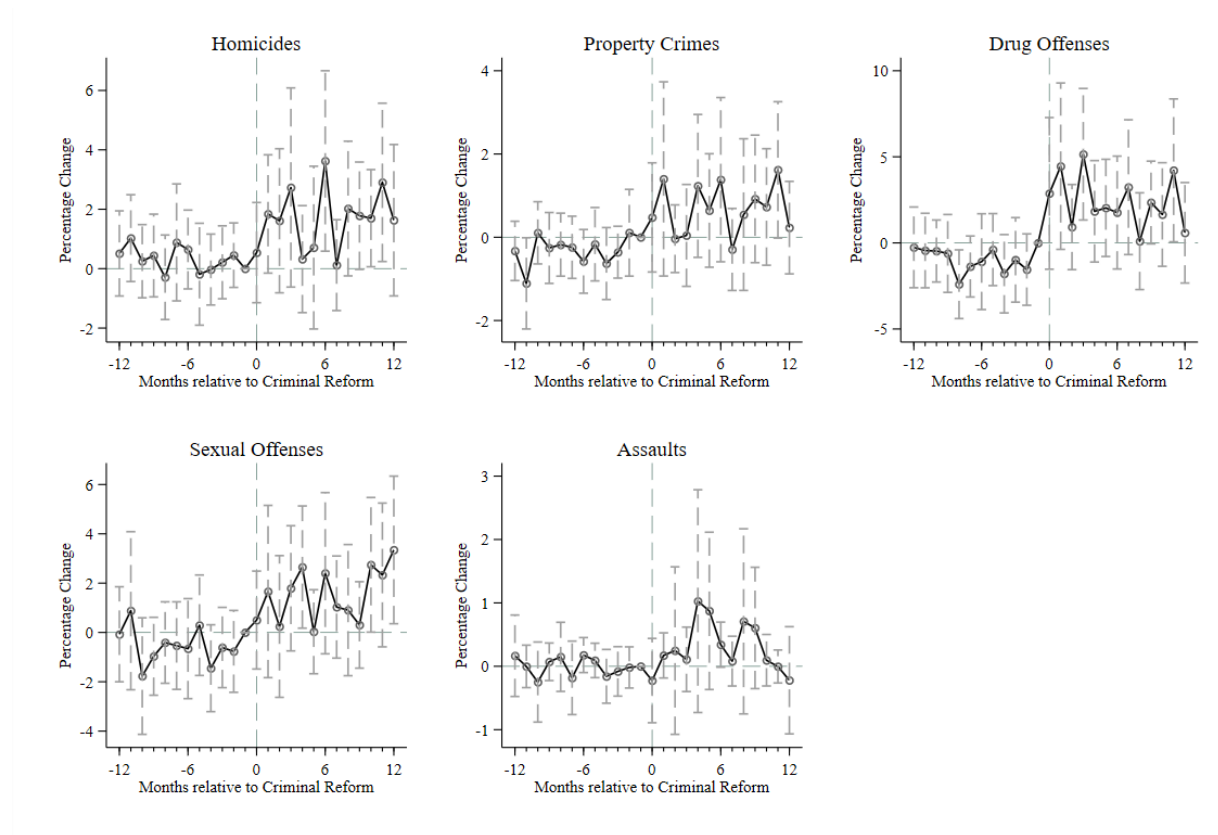
Note: this table shows the results of a two-way fixed effects regression of the days between different stages of the criminal procedure on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in each municipality and month (panel A) and the non-negative difference between a given month and the month of implementation up to 12 months (panel B). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, density of population, rural index, displaced population, and the lag of police arrest. Judicial district-month-clustered standard errors in parentheses. P-Wyong P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. *** p<0.01, ** p<0.05, * p<0.1

Figure 3. Leads-and-lags model: Jail-based pretrial detention



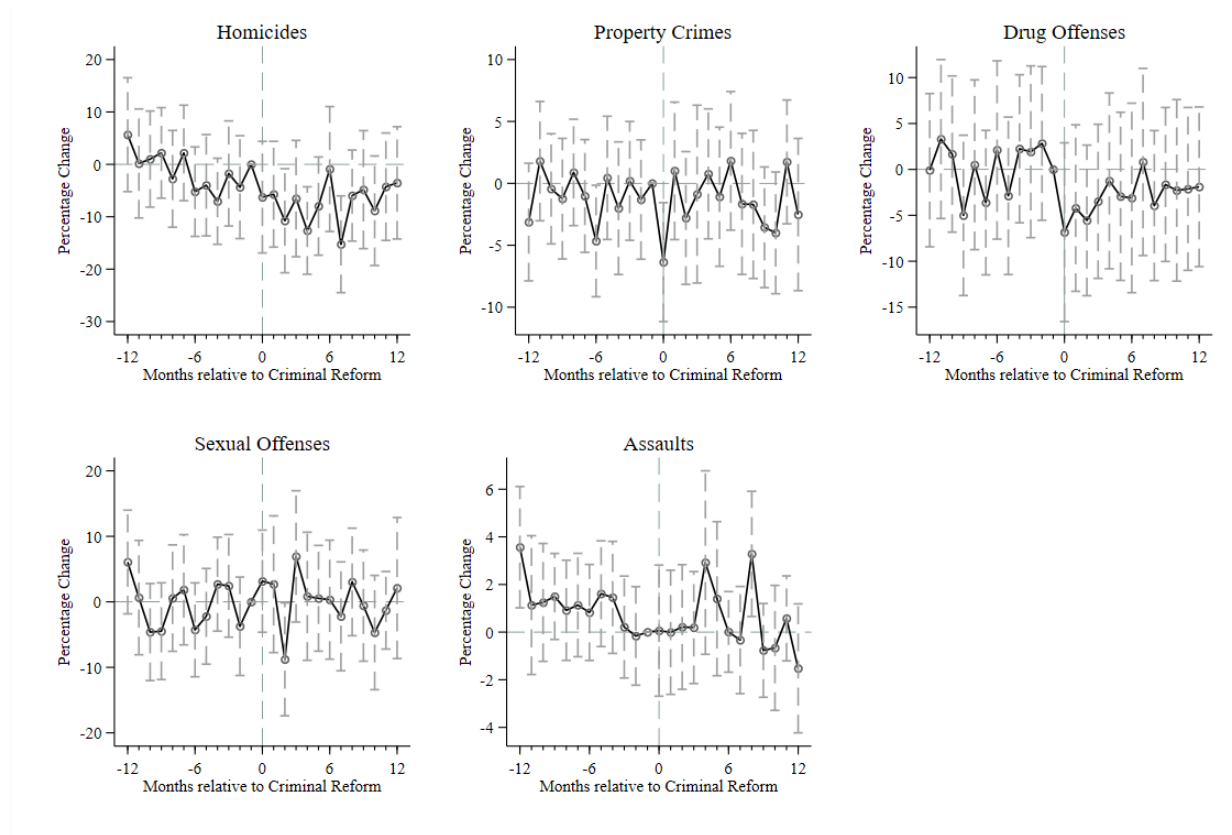
Note: this figure shows the results of an event study of jail-based pretrial detention rates for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Pretrial detention rates are computed as the ratio between the number of cases with active measures in a municipality and month relative to the total number of cases.

Figure 4. Leads-and-lags model: Pretrial house arrest



Note: this figure shows the results of an event study of the rates of pretrial house arrests for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Pretrial detention rates are computed as the ratio between the number of cases with active measures in a municipality and month relative to the total number of cases with imputations in that same municipality-month

Figure 5. Leads-and-lags model: Pretrial detention total



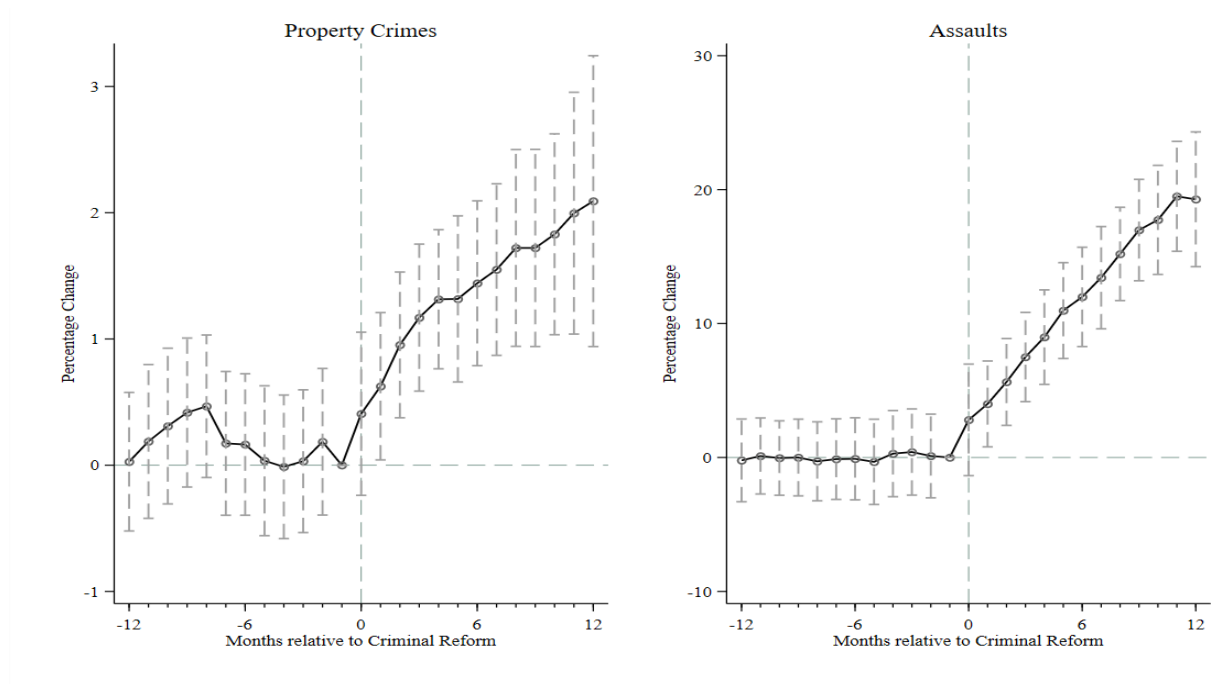
Note: this figure shows the results of an event study of the rates of total pretrial detentions for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Pretrial detention rates are computed as the ratio between the number of cases with active measures in a municipality and month relative to the total number of cases with imputations in that same municipality-month.

Table 3. Difference-in-Difference Results for Pretrial Detention

VARIABLES	(1) Homicides	(2) Property Crimes	(3) Drug Offenses	(4) Sexual Offenses	(5) Assaults
Panel A: Jail-based Pretrial Detention					
T	-7.48*** (1.44)	-1.43* (0.74)	-6.15*** (0.99)	-1.73 (1.22)	-1.06** (0.43)
R-squared	0.13	0.11	0.15	0.11	0.11
Mean if T==0	22.13	8.10	19.22	17.36	3.03
% Change after T	-33.8%	-17.7%	-32.0%	-9.9%	-34.9%
P-wyoming P-Value	0.00	0.88	0.00	0.20	0.97
Panel B: Pretrial Detention in House Arrests					
T	1.35*** (0.29)	0.91*** (0.23)	3.55*** (0.48)	1.93*** (0.37)	0.33** (0.14)
R-squared	0.14	0.10	0.10	0.12	0.07
Mean if T==0	0.15	0.19	1.34	0.67	0.07
% Change after T	859.0%	486.6%	266.0%	287.3%	465.2%
P-wyoming P-Value	0.00	0.01	0.00	0.00	0.33
Panel C: Total Pretrial Detention					
T	-6.13*** (1.47)	-0.52 (0.76)	-2.60** (1.16)	0.20 (1.27)	-0.73 (0.45)
R-squared	0.13	0.11	0.14	0.11	0.12
Mean if T==0	22.28	8.29	20.55	18.03	3.12
% Change after T	-27.5%	-6.3%	-12.6%	1.1%	-23.5%
P-wyoming P-Value	0.00	0.97	0.00	0.98	0.80
Observations	11,200	14,857	14,004	12,098	17,554
Year Month & Month-Year FE	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES

Note: this table shows the results of a two-way fixed effects regression of the rates of jail-based pretrial detentions (panel A), house arrests (panel B) and the total (panel C) for different crimes on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality. All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district - year-clustered standard errors in parentheses. P-Wyoming P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Pretrial detention rates are computed as the ratio of the number of cases with active measures in a municipality-month to the total number of cases with imputations in that same municipality-month. The percentage change after treatment was calculated $\frac{\text{Effect of T}}{\text{Mean T=0}} * 100$. *** p<0.01, ** p<0.05, * p<0.1.

Figure 6. Leads-and-lags model for Settlements rate



Note: this figure shows the results of an event study of settlements rates for the two minor crimes in our data for which a settlement hearing before the imputation of charges hearing is mandatory under the new system (property crimes and assault) as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Settlement rates are computed as the ratio between settlements and open cases for a specific crime in a municipality allowing for information delays and timespans, i.e., $Settlement\ Rate_{i,t}^s =$

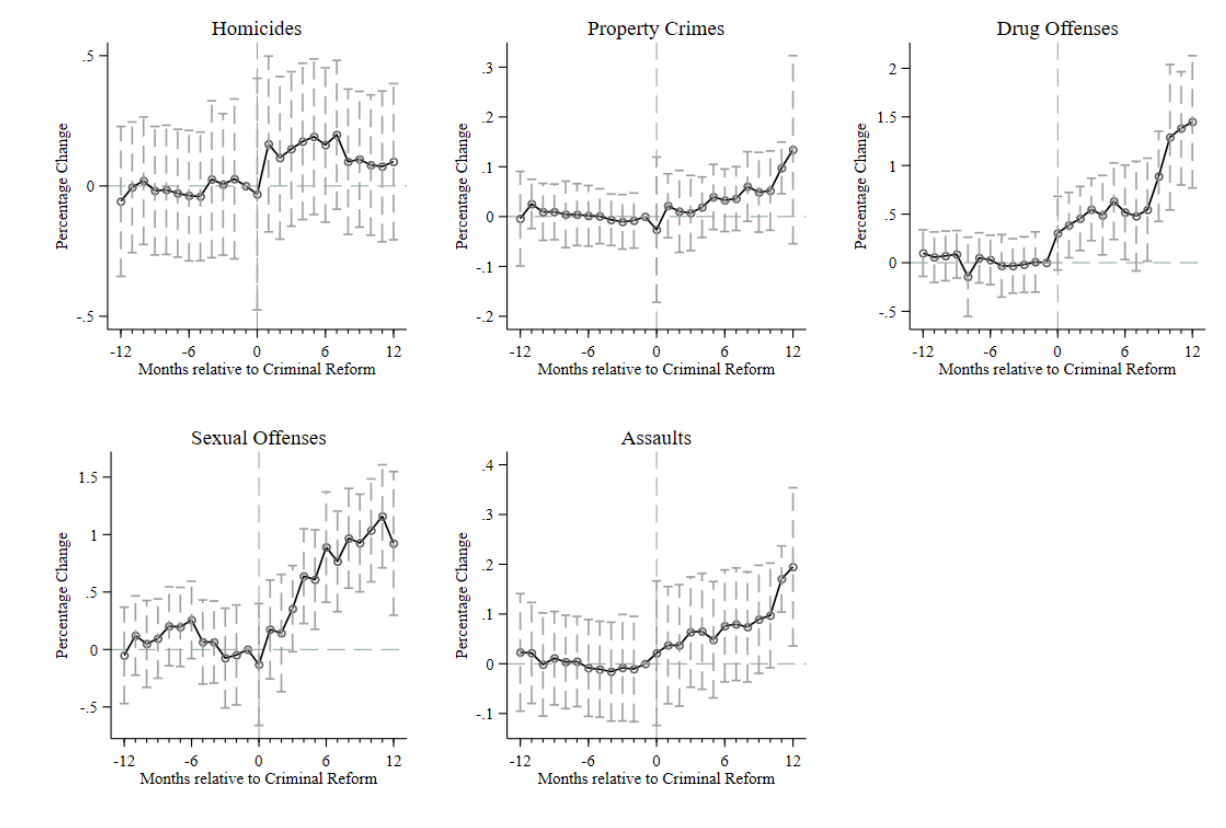
$$\frac{\sum_{t=0}^{-11} Settlements_{i,t}^s}{\sum_{t=-1}^{11} Open\ Cases_{i,t}^s}$$

Table 4. Difference-in-Difference Results for Settlements (Pre-Imputation)

VARIABLES	Panel A		Panel B	
	(1) Property Crimes	(2) Assaults	(3) Property Crimes	(4) Assaults
T	1.07*** (0.12)	10.31*** (0.64)	1.08*** (0.12)	10.51*** (0.58)
Exposure Time to T			0.03*** (0.01)	0.52*** (0.04)
Observations	32,064	32,035	32,064	32,035
R-squared	0.40	0.59	0.40	0.61
Year Month & Month-Year FE	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES
Controls	YES	YES	YES	YES
Mean if T=0	2.57	15.78	2.57	15.78
% Change after T	41.7%	65.3%	54.24%	105.89%
P-wyoming P-Value	0.00	0.00	0.00	0.00

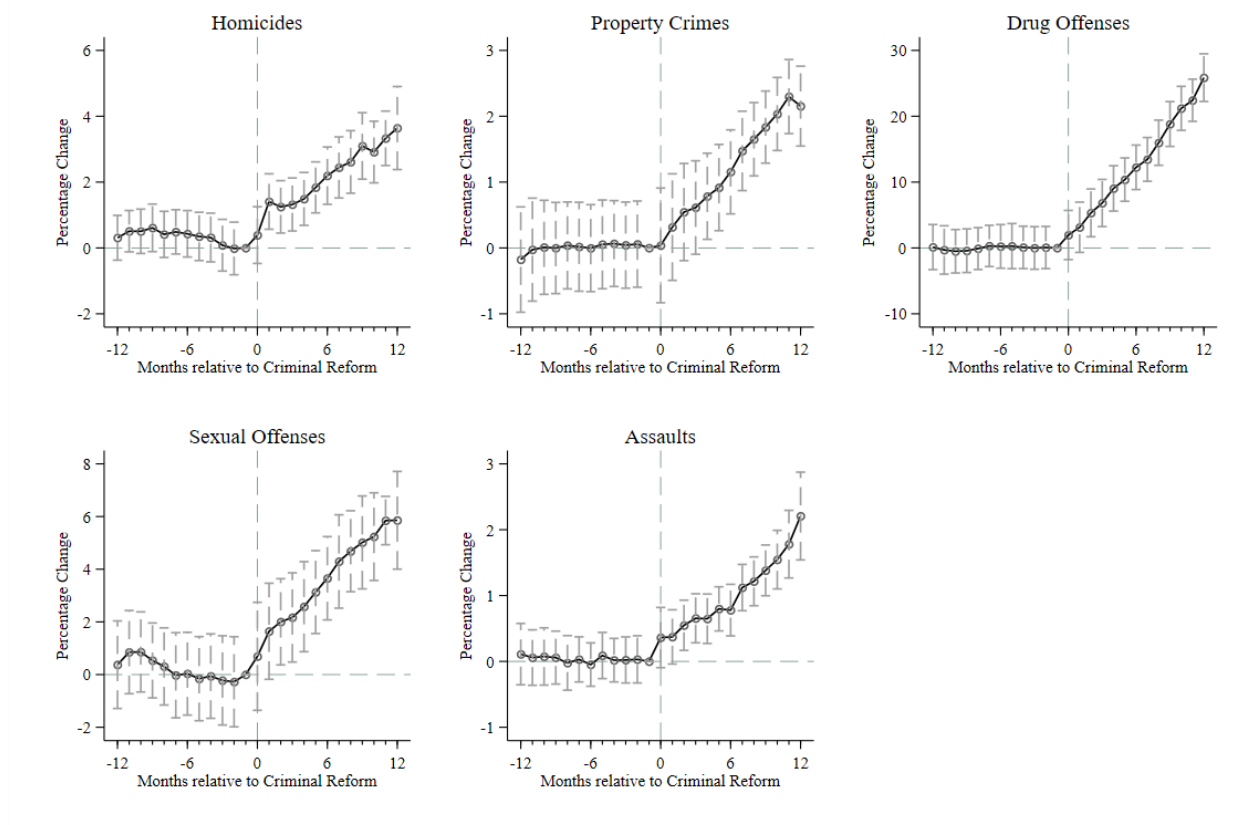
Note: this table shows the results of a two-way fixed effects regression of settlement rates for different crimes on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (panel A) and the non-negative difference between a given month and the month of implementation up to 12 months (panel B). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, density of population, rural index, displaced population, and the lag of police arrest. Judicial district-year-clustered standard errors in parentheses. P-Wyoming P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Settlement rates are computed as the ratio between settlements and open cases for a specific crime in a municipality allowing for information delays and timespans, i.e., $Settlement\ Rate_{i,t}^s = \frac{\sum_{t=0}^{11} Settlements_{i,t}^s}{\sum_{t=-1}^{11} Open\ Cases_{i,t}^s}$. The percentage change after treatment was calculated $\frac{Effect\ of\ T}{Mean\ T=0} * 100$ (panel A). To estimate the percentage change after treatment of panel B, we calculated the effect after one month of implementation as $\frac{\beta_1 + \beta_2 * 12}{Mean\ T=0} * 100$. ***. *** p<0.01, ** p<0.05, * p<0.1.

Figure 7A. Leads-and-lags model for acquittal rates



Note: this figure shows the results of an event study of acquittals rates for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Rates are computed as the ratio between the total number of acquittals and open cases for a specific crime in a municipality allowing for information delays and timespans, i.e., $Acquittals\ Rate_{i,t}^s = \frac{\sum_{t=0}^{-11} Acquittals_{i,t}^s}{\sum_{t=-1}^{11} Open\ Cases_{i,t}^s}$.

Figure 7B. Leads-and-lags model for conviction rates



Note: this figure shows the results of an event study of convictions rates for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed Judicial district-month-clustered standard errors. Rates are computed as the ratio between the total number of convictions and open cases for a specific crime in a municipality allowing for information delays and timespans, i.e.,

$$Convictions\ Rate_{i,t}^S = \frac{\sum_{t=0}^{-11} Convictions_{i,t}^S}{\sum_{t=-1}^{11} Open\ Cases_{i,t}^S}.$$

Table 5. Difference-in-Difference Results for Total Acquittal and Conviction Rates

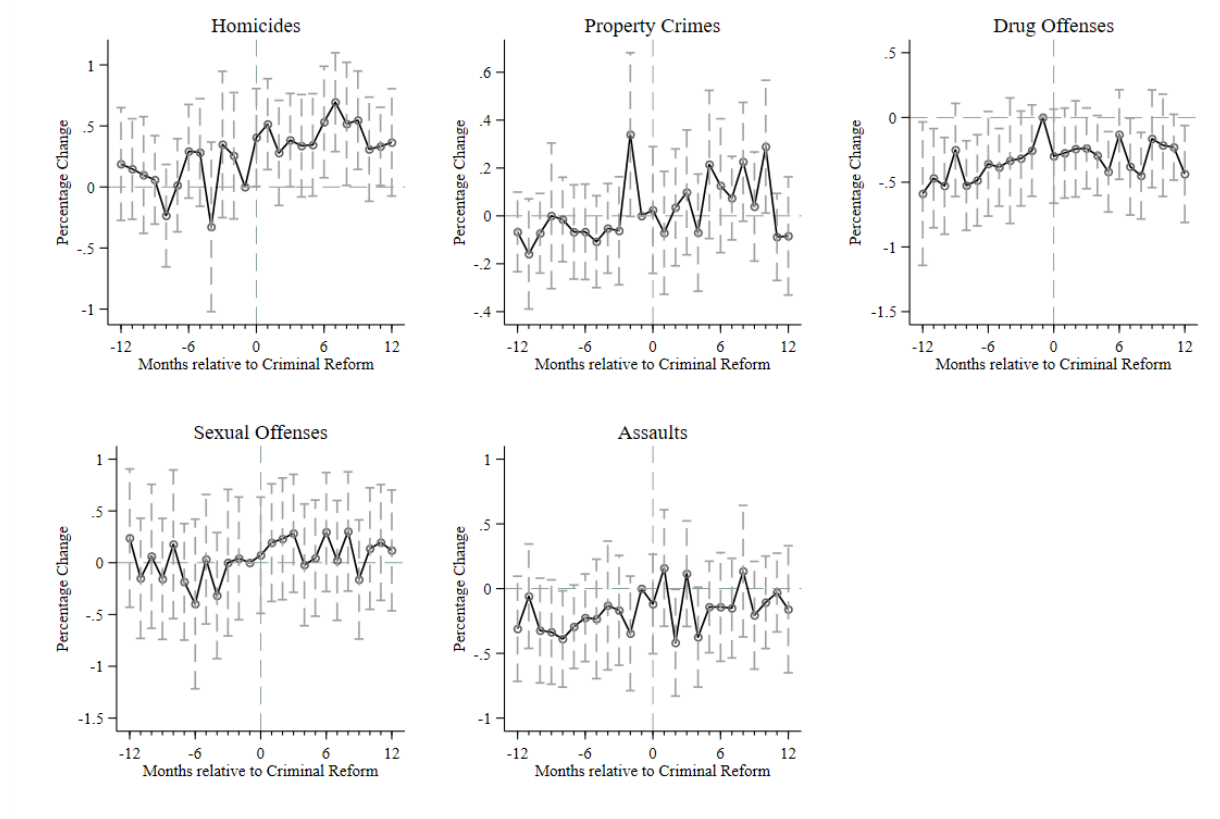
VARIABLES	(1) Homicides	(2) Property Crimes	(3) Drug Offenses	(4) Sexual Offenses	(5) Assaults
Total Acquittals					
Panel A					
T	0.12*** (0.04)	0.01 (0.01)	0.59*** (0.08)	0.51*** (0.07)	0.06*** (0.01)
R-squared	0.36	0.37	0.47	0.40	0.22
% Change after T	407.6%	106.0%	1617.1%	1988.3%	301.5%
P-wyyoung P-Value	0.00	0.00	0.00	0.00	0.00
Panel B					
T	0.13*** (0.04)	0.01 (0.01)	0.61*** (0.07)	0.54*** (0.07)	0.06*** (0.01)
Exposure Time to T	0.02*** (0.00)	0.01*** (0.00)	0.02*** (0.00)	0.06*** (0.01)	0.01*** (0.00)
R-squared	0.36	0.37	0.47	0.40	0.22
% Change after T	1098.4%	1155.9%	2261.7%	4719.8%	799.0%
P-wyyoung P-Value	0.04	0.60	0.00	0.01	0.06
Mean if T=0	0.03	0.01	0.04	0.03	0.02
Total Convictions					
Panel A					
T	1.44*** (0.13)	0.97*** (0.01)	9.69*** (0.71)	2.88*** (0.25)	0.77*** (0.08)
R-squared	0.40	0.42	0.54	0.39	0.32
% Change after T	885.9%	1053.0%	2257.8%	687.1%	707.3%
P-wyyoung P-Value	0.00	0.00	0.00	0.00	0.00
Panel B					
T	1.49*** (0.12)	1.01*** (0.10)	10.43*** (0.65)	3.01*** (0.23)	0.79*** (0.08)
Exposure Time to T	0.12*** (0.01)	0.09*** (0.01)	0.74*** (0.03)	0.25*** (0.01)	0.04*** (0.00)
R-squared	0.41	0.44	0.57	0.40	0.32
% Change after T	1803.1%	2204.3%	4488.8%	1429.1%	1127.5%
P-wyyoung P-Value	0.00	0.00	0.00	0.00	0.00
Mean if T=0	0.16	0.09	0.43	0.42	0.11
Observations	31,760	32,064	26,267	31,169	32,035
Year Month & Month-Year FE	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES

Note: this table shows the results of a two-way fixed effects regression of acquittals (top panel) or conviction rates (bottom panel) for different crimes on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (panel A) and the non-negative difference between a given month and the month of implementation up to 12 months (panel B). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district - year-clustered standard errors in parentheses. P-Wyyoung P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Rates are computed as the ratio between the total number of acquittals or convictions and open criminal complaints for a specific crime in a municipality allowing for information delays and timespans, i.e.,

$Acquittals\ or\ Convictions\ Rate_{i,t}^S = \frac{\sum_{t=0}^{-11} Acquittals\ or\ Convictions_{i,t}^S}{\sum_{t=-1}^{11} Open\ Criminal\ Complaints_{i,t}^S}$. The percentage change after treatment was calculated

$\frac{Effect\ of\ T}{Mean\ T=0} * 100$ (panel A). To estimate the percentage change after treatment of panel B, we calculated the effect after one month of implementation as $\frac{\beta_1 + \beta_2 * 12}{Mean\ T=0} * 100$. *** p<0.01, ** p<0.05, * p<0.1

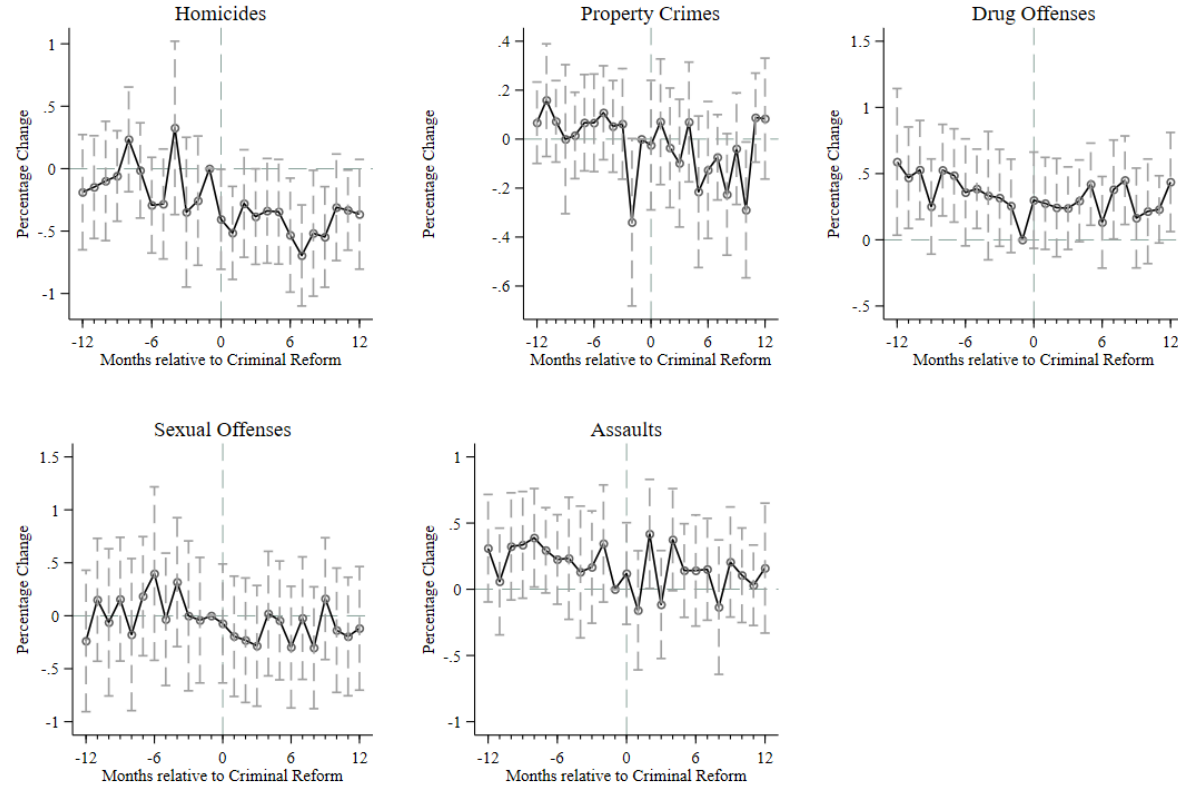
Figure 8A. Leads-and-lags model for the share of acquittals in trial over total sentences in trial



Note: this figure shows the results of an event study of the in-trial acquittals for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Rates are computed as the ratio between the number of in-trial acquittals and open cases for a specific crime in a municipality allowing for information delays and timespans, i.e.,

$$Acquittals\ Rate_{i,t}^s = \frac{\sum_{t=-11}^{-1} Acquittals_{i,t}^s}{\sum_{t=-11}^{-1} Open\ Cases_{i,t}^s}.$$

Figure 8B. Leads-and-lags model for the share of convictions in trial over total sentences in trial



Note: this figure shows the results of an event study of the in-trial convictions for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Rates are computed as the ratio between the number of in-trial convictions and open cases for a specific crime in a municipality, allowing for information delays and timespans, i.e.,

$$Convictions\ Rate_{i,t}^s = \frac{\sum_{t=0}^{-11} Convictions_{i,t}^s}{\sum_{t=-1}^{11} Open\ Cases_{i,t}^s}.$$

Table 6. Difference-in-Difference results for the share of in-trial acquittals and conviction over total in-trial sentences

VARIABLES	(1) Homicides	(2) Property Crimes	(3) Drug Offenses	(4) Sexual Offenses	(5) Assaults
Acquittals in Court					
Panel A					
T	0.28*** (0.08)	0.11** (0.05)	0.11* (0.07)	0.16* (0.08)	0.12* (0.07)
R-squared	0.47	0.39	0.52	0.44	0.47
% Change after T	216.9%	114.6%	150.6%	153.5%	89.9%
P-wyyoung P-Value	0.01	0.01	0.00	0.00	0.58
Panel B					
T	0.26*** (0.08)	0.09* (0.05)	0.14** (0.07)	0.14* (0.08)	0.13* (0.08)
Exposure Time to T	-0.00 (0.00)	-0.00 (0.00)	0.01 (0.00)	-0.00 (0.00)	0.00 (0.00)
R-squared	0.47	0.39	0.52	0.44	0.47
% Change after T	173.8%	43.9%	268.7%	118.8%	123.2%
P-wyyoung P-Value	0.01	0.01	0.01	0.00	0.51
Mean if T==0	0.13	0.09	0.07	0.10	0.14
Convictions in Court					
Panel A					
T	-0.28*** (0.08)	-0.11** (0.05)	-0.11* (0.07)	-0.16* (0.08)	-0.12* (0.07)
R-squared	0.47	0.39	0.52	0.44	0.47
% Change after T	-32.4%	-11.8%	-12.0%	-17.2%	-14.4%
P-wyyoung P-Value	0.01	0.01	0.00	0.00	0.58
Panel B					
T	-0.26*** (0.08)	-0.09* (0.05)	-0.14** (0.07)	-0.14* (0.08)	-0.13* (0.08)
Exposure Time to T	0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)
R-squared	0.47	0.39	0.52	0.44	0.47
% Change after T	-26.0%	-4.5%	-21.4%	-13.3%	-19.7%
P-wyyoung P-Value	0.01	0.01	0.01	0.00	0.51
Mean if T==0	0.87	0.91	0.93	0.90	0.86
Observations	1,068	1,314	1,002	1,431	947
Year Month & Month-Year FE	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES

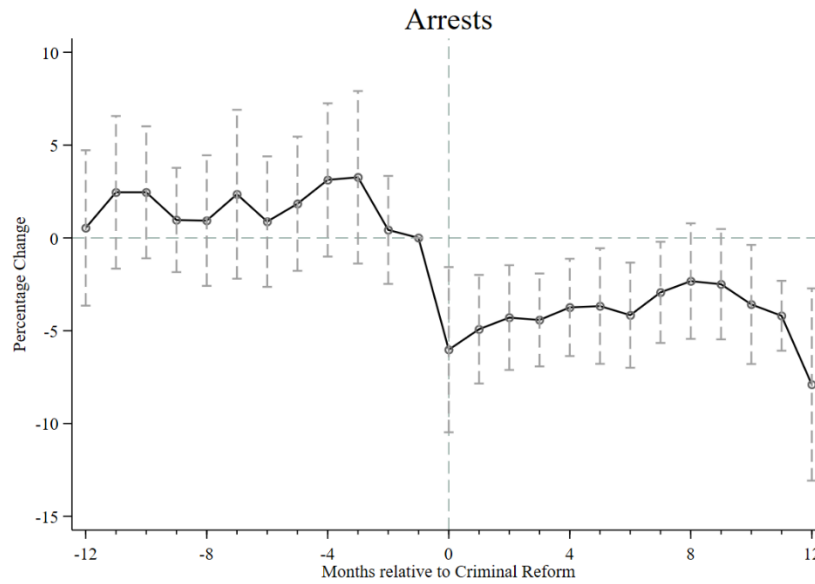
Note: this table shows the results of a two-way fixed effects regression of the in-trial acquittals (top panel) or conviction rates (bottom panel) for different crimes on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (panel A) and the non-negative difference between a given month and the month of implementation up to 12 months (panel B). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. P-Wyyoung P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Judicial district-year-clustered standard errors in parentheses. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Rates are computed as the ratio between the number of in-trial acquittals or convictions and total in-trial decisions for a specific crime in a municipality allowing for information delays and timespans, i.e.,

$Court\ Acquittals\ or\ Convictions\ Rate_{i,t}^s = \frac{\sum_{t=0}^{11} In-trial\ Acquittals\ or\ Convictions_{i,t}^s}{\sum_{t=-1}^{11} In-trial\ decisions_{i,t}^s}$. The percentage change after treatment

was calculated $\frac{Effect\ of\ T}{Mean\ T=0} * 100$ (panel A). To estimate the percentage change after treatment of panel B, we calculated

the effect after one month of implementation as $\frac{\beta_1 + \beta_2 * 12}{Mean\ T=0} * 100$. *** p<0.01, ** p<0.05, * p<0.1

Figure 9. Leads-and-lags model for arrest rates



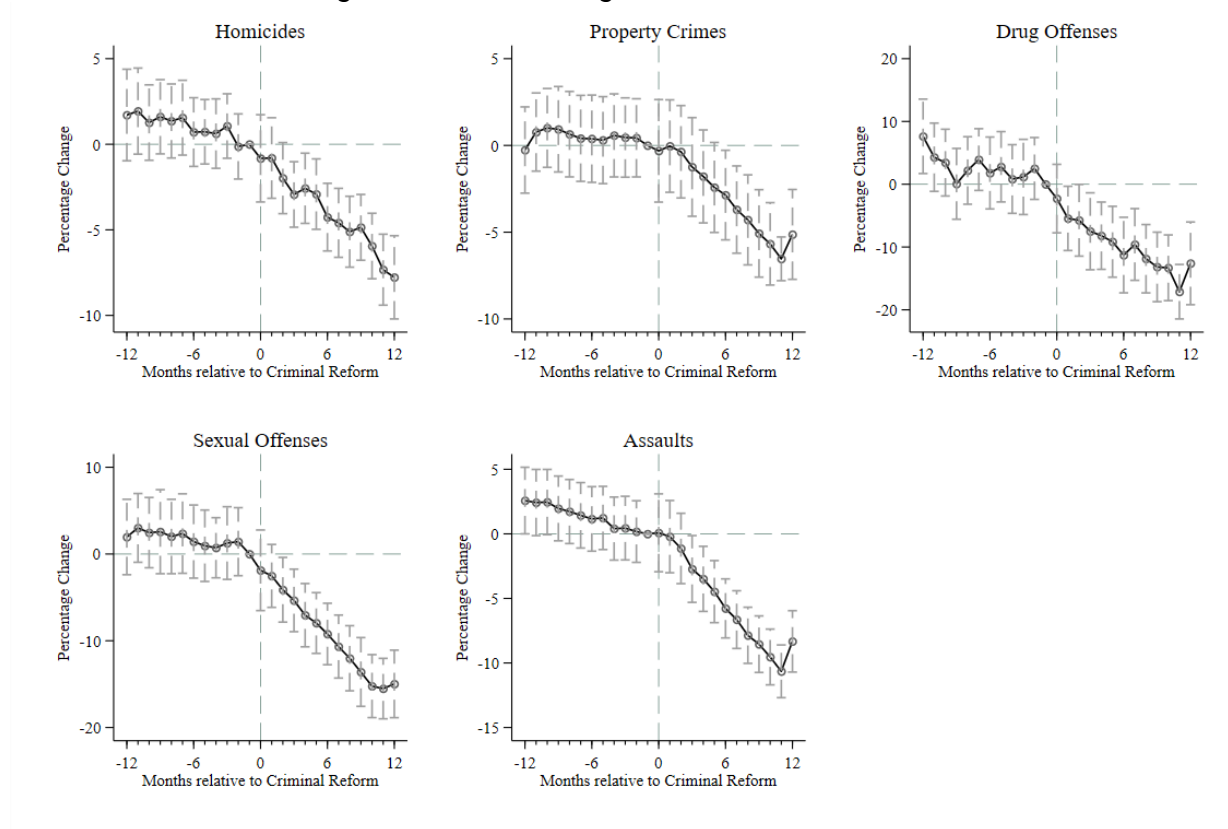
Note: this figure shows the results of an event study of the arrest rate by 100,000 inhabitants as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with judicial district-month-clustered standard errors.

Table 7. Difference-in-Difference results for arrest rates

VARIABLES	(1)	(2)
	Arrest rate	Arrest rate
T	-5.46*** (0.54)	-5.10*** (0.49)
Exposure Time to T		-0.19*** (0.05)
Observations	78,894	77,094
R-squared	0.32	0.31
Year Month & Month-Year FE	YES	YES
Municipio FE	YES	YES
Controls	YES	YES
Mean T = 0	14.05	14.05
Effect of T	-38.9%	-52.4%

Note: this table shows the results of a two-way fixed effects regression of the logarithm of arrest rates on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (column 1) and the non-negative difference between a given month and the month of implementation up to 12 months (column 2). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district-year-clustered standard errors in parentheses. P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Effect of T is calculated as $\frac{\text{Effect of T}}{\text{Mean T}=0} * 100$ for column (1) and as $\frac{\beta_1 + \beta_2 * 12}{\text{Mean T}=0} * 100$ for column (2).

Figure 10. Leads-and-lags model for clearance rates



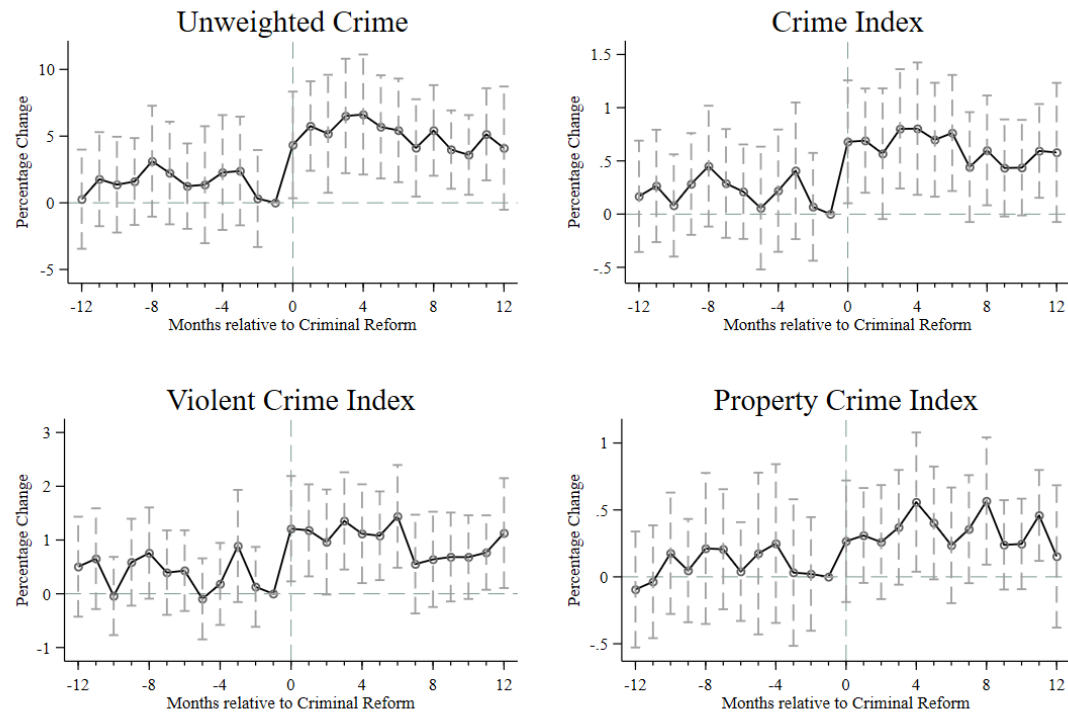
Note: this figure shows the results of an event study of the logarithm of clearance rates for different crimes as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Clearance rates are computed as the ratio between imputations and open cases for a specific crime in a municipality, allowing for information delays and timespans, i.e.,

$$Clearance\ Rate_{i,t}^s = \frac{\sum_{t=0}^{-11} Imputations_{i,t}^s}{\sum_{t=-1}^{11} Open\ Cases_{i,t}^s}.$$

Table 8. Difference-in-Difference results for Clearance rates

VARIABLES	(1) Homicides	(2) Property Crimes	(3) Drug Offenses	(4) Sexual Offenses	(5) Assaults
Panel A					
T	-4.01*** (0.34)	-3.27*** (0.29)	-11.06*** (0.79)	-9.64*** (0.59)	-5.99*** (0.38)
% Change after T	-23.3%	-24.2%	-15.6%	-24.7%	-27.3%
R-squared	0.39	0.48	0.42	0.43	0.60
P-wyong P-Value	0.00	0.00	0.00	0.00	0.00
Panel B					
T	-3.98*** (0.35)	-3.23*** (0.30)	-10.71*** (0.77)	-9.60*** (0.60)	-6.00*** (0.38)
Exposure Time to T	0.08*** (0.02)	0.13*** (0.03)	0.35*** (0.06)	0.07 (0.05)	-0.02 (0.03)
% Change after T	-17.2%	-12.8%	-9.2%	-22.3%	-28.3%
R-squared	0.385	0.484	0.418	0.431	0.60
P-wyong P-Value	0.00	0.00	0.00	0.00	0.00
Observations	31,760	32,064	26,267	31,169	32,035
Year Month & Month-Year FE	YES	YES	YES	YES	YES
Municipality FE	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES
Mean if T==0	17.21	13.50	70.74	39.03	21.94
Note: this table shows the results of a two-way fixed effects regression of the clearance rates for different crimes on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (panel A) and the non-negative difference between a given month and the month of implementation up to 12 months (panel B). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district-year-clustered standard errors in parentheses. P-Wyong P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Clearance rates are computed as the ratio between imputations and open criminal complaints for a specific crime in a municipality allowing for information delays and time spans, i.e., $Clearance\ Rate_{i,t}^s = \frac{\sum_{t=0}^{11} Imputations_{i,t}^s}{\sum_{t=-1}^{11} Open\ Criminal\ Complaints_{i,t}^s}$. The percentage change after treatment was calculated $\frac{Effect\ of\ T}{Mean\ T=0} * 100$. To estimate the percentage change after treatment of panel B, we calculated the effect after one month of implementation as $\frac{\beta_1 + \beta_2 * 12}{Mean\ T=0} * 100$. *** p<0.01, ** p<0.05, * p<0.1					

Figure 11. Leads-and-lags model for Aggregate Crime Indices



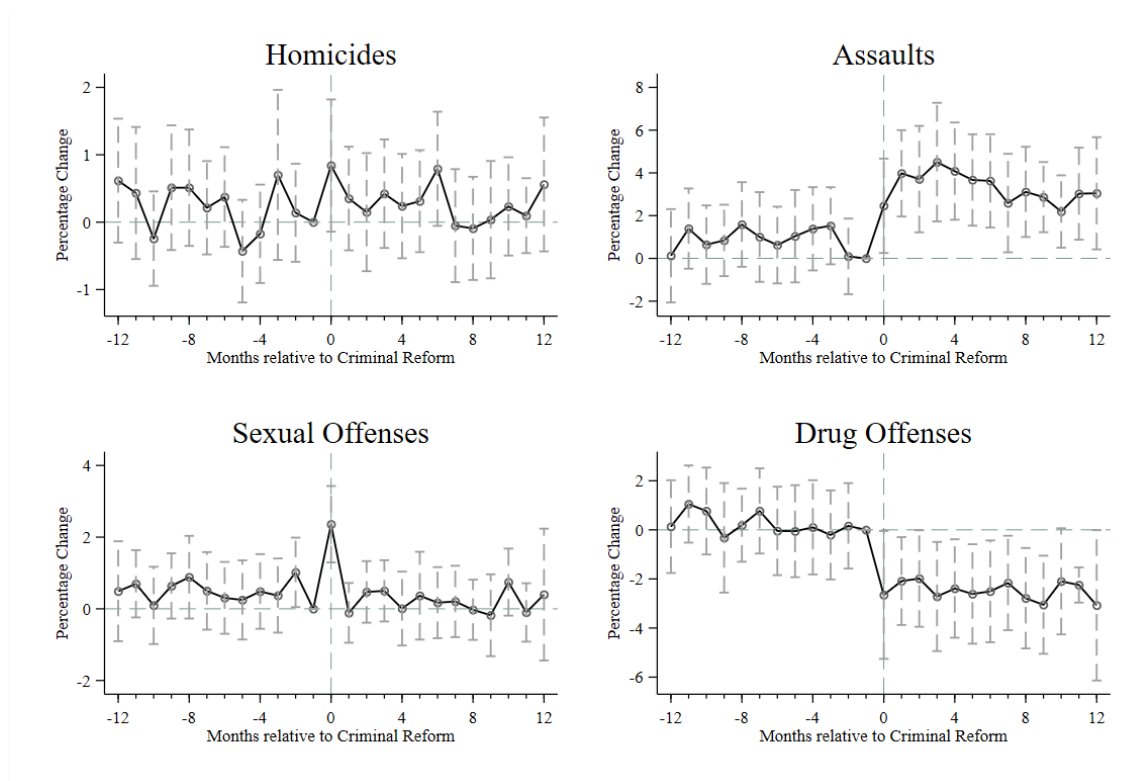
Note: this figure shows the results of an event study of the logarithm of different crime rates as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors. Crime indices are computed as the weighted average of different types of crimes, where the weights are given by the average sentence length of each type of crime.

Table 9. Difference-in-Difference results for Aggregate Crime Indices

VARIABLES	Panel A				Panel B			
	(1) Unweighted Crime	(2) Crime Index	(3) Violent Crime Index	(4) Property Crime Index	(5) Unweighted Crime	(6) Crime Index	(7) Violent Crime Index	(8) Property Crime Index
T	4.05*** (0.60)	0.46*** (0.08)	0.62*** (0.13)	0.33*** (0.08)	3.56*** (0.60)	0.41*** (0.08)	0.61*** (0.13)	0.25*** (0.08)
Exposure Time to T					0.21*** (0.03)	0.02*** (0.00)	0.01 (0.01)	0.04*** (0.00)
Observations	75,976	75,976	75,976	75,976	75,976	75,976	75,976	75,976
R-squared	0.30	0.22	0.15	0.35	0.30	0.22	0.15	0.35
P-wyoming P-Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Municipio FE	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Mean T = 0	13.53	2.45	4.12	1.16	13.53	2.45	4.12	1.16
Effect of T	30.0%	18.7%	15.2%	28.7%	44.5%	27.4%	16.3%	57.7%

Note: this table shows the results of a two-way fixed effects regression of different crime rates on an indicator variable that equals one if the adversarial procedural reform has been implemented in a given municipality and month (all columns) and the non-negative difference between a given month and the month of implementation up to 12 months (columns 5 to 8). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district-month-clustered standard errors in parentheses. P-Wyoming P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Effect of T is calculated $\frac{\text{Effect of T}}{\text{Mean T=0}} * 100$. for columns (1) to (4) and as $\frac{\beta_1 + \beta_2 * 12}{\text{Mean T=0}} * 100$ for columns (5) to (8). Crime indices are computed as the weighted average of different types of crimes, where the weights are given by the average sentence length of each type of crime and are presented in Table A2. *** p<0.01, ** p<0.05, * p<0.1

Figure 12A. Leads-and-lags model for Violent and Drug-related crimes



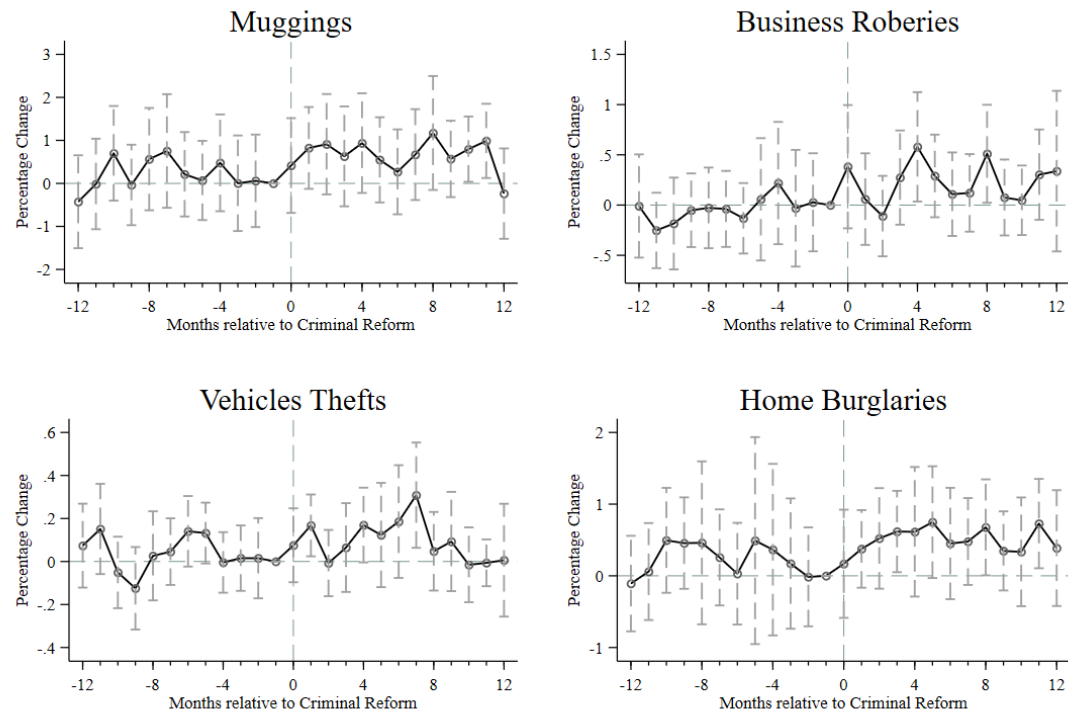
Note: this figure shows the results of an event study of different crime rates as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors.

Table 10A. Difference-in-Difference results for Violent and Drug-related crimes

VARIABLES	Panel A				Panel B			
	(1) Homicides	(2) Assaults	(3) Sexual Offenses	(4) Drug Offenses	(5) Homicides	(6) Assaults	(7) Sexual Offenses	(8) Drug Offenses
T	0.04 (0.12)	2.58*** (0.33)	-0.09 (0.16)	-2.76*** (0.25)	0.07 (0.19)	2.43*** (0.32)	-0.12 (0.16)	-3.14*** (0.27)
Exposure Time to T					-0.01 (0.01)	0.06*** (0.02)	0.02* (0.01)	0.21*** (0.03)
Observations	75,976	75,976	60,882	60,882	75,976	75,976	60,882	60,882
R-squared	0.14	0.17	0.13	0.30	0.14	0.17	0.13	0.31
P-wyong P-Value	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Year Month & Month-Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Municipio FE	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Mean T = 0	3.98	4.52	4.13	2.94	3.98	4.52	4.13	2.94
Effect of T	1.1%	57.0%	-2.1%	-93.9%	-1.8%	70.0%	1.8%	-21.6%

Note: this table shows the results of a two-way fixed effects regression of different crime rates on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (all columns) and the non-negative difference between a given month and the month of implementation up to 12 months (columns 5 to 8). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district - year-clustered standard errors in parentheses. P-Wyong P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Effect of T is calculated as $\frac{\beta_1}{\text{Mean } T = 0} * 100$ for columns (1) to (4) and as $\frac{\beta_1 + \beta_2 * 12}{\text{Mean } T = 0} * 100$ for columns (5) to (8). *** p<0.01, ** p<0.05, * p<0.1

Figure 12B. Leads-and-lags model for Property crimes



Note: this figure shows the results of an event study of the logarithm of different crime rates as a function of the leads and lags relative to the month of implementation of the reform in a municipality. Dashed lines represent 95% confidence intervals computed with Judicial district-month-clustered standard errors.

Table 10B. Difference-in-Difference results for Property crimes

VARIABLES	Panel A				Panel B			
	(1) Muggings	(2) Business Robberies	(3) Vehicles Thefts	(4) Home Burglaries	(5) Muggings	(6) Business Robberies	(7) Vehicles Thefts	(8) Home Burglaries
T	0.73*** (0.23)	0.26*** (0.08)	0.06* (0.03)	0.37*** (0.13)	0.47** (0.24)	0.21** (0.08)	0.07** (0.04)	0.30** (0.13)
Exposure Time to T					0.11*** (0.01)	0.02*** (0.00)	-0.00* (0.00)	0.03*** (0.01)
Observations	75,976	75,976	75,976	75,976	75,976	75,976	75,976	75,976
R-squared	0.30	0.14	0.11	0.19	0.31	0.14	0.11	0.19
P-wyoung P-Value	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
Year Month & Month- Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Municipio FE	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Mean T = 0	2.42	0.78	0.44	1.39	2.42	0.78	0.44	1.39
Effect of T	30.3%	34.1%	14.6%	26.6%	73.5%	62.5%	8.2%	45.9%

Note: this table shows the results of a two-way fixed effects regression of different crime rates on an indicator variable that equals 1 if the adversarial procedural reform has been implemented in a given municipality and month (all columns) and the non-negative difference between a given month and the month of implementation up to 12 months (columns 5 to 8). All regressions control for municipality, year, month, and year*month fixed effects, as well as for per capita Industry and Business tax collection, per capita investment in education, fiscal performance, population density, rural index, displaced population, and the lagged police arrest rate. Judicial district-month-clustered standard errors in parentheses. P-Wyong P-Value: P-value adjusted using Young's method (1990), used as a multiple hypothesis testing procedure to correct for dependence in errors. Mean T=0 corresponds to the mean of the dependent variable for those observations in the control group before the implementation of the reform. Effect of T is calculated as $\frac{\beta_1}{Mean\ T=0} * 100$ for columns (1) to (4) and as $\frac{\beta_1 + \beta_2 * 12}{Mean\ T=0} * 100$ for columns (5) to (8). *** p<0.01, ** p<0.05, * p<0.1